Inspection of Pressure–Relieving Devices

API RECOMMENDED PRACTICE 576
FIRST EDITION, SEPTEMBER 1992

American Petroleum Institute
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Inspection of Pressure–Relieving Devices

Refining Department

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FOREWORD

This recommended practice covers the material previously presented as Chapter XVI of the API *Guide for Inspection of Refinery Equipment*. The material originally published in the *Guide* is being reorganized to be published in several recommended practices.

This recommended practice is based on the accumulated knowledge and experience of engineers and other personnel in the petroleum industry. The information in this recommended practice does not constitute and should not be construed as a code of rules, regulations, or minimum safe practices. The practices described in this publication are not intended to supplant other practices that have proven satisfactory, nor is this publication intended to discourage innovation and originality in the inspection of refineries. Users of this recommended practice are reminded that no book or manual is a substitute for the judgment of a responsible, qualified person.

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CONTENTS

	Page
SECTION 1—GENERAL	1
1.1 Scope	1
1.2 Referenced Publications	1
SECTION 2—PRESSURE-RELIEVING DEVICES	1
2.1 General	1
2.2 Types of Pressure-Relieving Devices	2
2.2.1 Pressure Relief Valve	2
2.2.2 Pressure and/or Vacuum Vent Valve	7
2.2.3 Rupture Disk Device	7
2.3 Variations With Resilient Valve Seats	11
2.3.1 General	11
2.3.2 Applications	11
2.3.3 Limitations	11
SECTION 3—REASONS FOR INSPECTION	13
3.1 To Protect Personnel and Equipment	13
3.2 To Determine the Condition of a Device	13
3.3 To Evaluate Inspection Frequency	13
SECTION 4—CAUSES OF IMPROPER PERFORMANCE	13
4.1 Corrosion	13
4.2 Damaged Seating Surfaces	13
4.3 Failed Springs	15
4.4 Improper Setting and Adjustment	16
4.5 Plugging and Sticking	18
4.6 Misapplication of Materials	19
4.7 Improper Location, History, or Identification	19
4.8 Rough Handling	19
4.8.1 General	19
4.8.2 During Shipment	20
4.8.3 During Maintenance	20
4.8.4 During Installation	20
4.9 Improper Differential Between Operating and Set Pressures	20
4.10 Improper Piping Test Procedures	21
SECTION 5—FREQUENCY AND TIME OF INSPECTION	21
5.1 Inspection Frequency	21
5.1.1 General	21
5.1.2 Normal Basis	21
5.1.3 Manufacturer's Basis	21
5.1.4 Other Bases	21
5.2 Time of Inspection	21
5.2.1 General	21
5.2.2 Inspection on New Installations	22
5.2.3 Unscheduled Inspection	22
5.2.4 Inspection During Planned Shutdowns	22
5.2.5 Inspection After Extended Shutdowns	22
5.2.6 Visual On-Stream Inspection	23
SECTION 6—PREPARATION, TOOLS, AND EQUIPMENT	24
6.1 Preparation	24
6.1.1 Equipment Checked for Safety	24

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6.1.2 Valve Identification Correctly Installed 24
6.1.3 Operating Conditions Noted 25
6.1.4 Careful and Minimal Removal of Device From System In Operation 25
6.2 Shop Test Block 25
6.2.1 Need and Function of Test Block 25
6.2.2 Testing With Air 26
6.2.3 Testing With Water 26
6.2.4 Description of Test Block 26
6.3 Tools and Spare Parts 26
6.3.1 Tools 26
6.3.2 Spare Parts 29

SECTION 7—INSPECTION, TRANSPORTATION, TESTING, MAINTENANCE, AND SETTING 29
7.1 Initial Inspection of Pressure Relief Valves and Adjacent Piping and Transportation of Valves to Shop 29
7.1.1 Inspection of Valves 29
7.1.2 Inspection of Adjacent Inlet and Outlet Piping 30
7.1.3 Transportation of Valves to Shop 30
7.2 Inspection, Testing, Maintenance, and Setting of Pressure Relief Valves in the Shop 30
7.2.1 Pop Testing of Valve "As Received" 30
7.2.2 Visual Inspection 30
7.2.3 Determining "As Received" Relieving Pressure 30
7.2.4 Dismantling of Valve 31
7.2.5 Cleaning and Inspection of Parts 31
7.2.6 Reconditioning and Replacement of Parts 31
7.2.7 Reassembly of Valve 31
7.2.8 Setting of Valve to Set Pressure 31
7.2.9 Checking of Valve for Tightness 32
7.2.10 Completion of Necessary Records 32
7.3 Inspection, Testing, Maintenance, and Setting of Pressure Relief Valves on Equipment 32
7.3.1 Process Units 32
7.3.2 Storage Tanks 32
7.3.3 Boilers 33
7.4 Inspection, Testing, Maintenance, and Setting of Pilot-Operated Safety Relief Valves 33
7.5 Inspection and Replacement of Rupture Disks 34

SECTION 8—RECORDS AND REPORTS 34
8.1 Objective 34
8.2 The Need to Keep Records 34
8.3 Responsibilities 35
8.4 Sample Record and Report System 35

APPENDIX—SAMPLE RECORD AND REPORT FORMS 37

Figures
1—Top-Guided Safety Valve 2
2—Relief Valve 3
3—Conventional Safety Relief Valve 4
4—Balanced Bellows Safety Relief Valve 4
5—Balanced Bellows Safety Relief Valve With Auxiliary Balancing Piston 5
6—Low-Pressure Diaphragm-Type Pilot-Operated Valve 6

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7—High-Pressure Pilot-Operated Valve 7
8—High-Pressure Pilot-Operated Valve With Optional Dual Outlets 8
9—Weight-Loaded Pallet Vent Valve 9
10—Spring- and Weight-Loaded Vent Valve 9
11—Rupture Disk 10
12—Rupture Disk Device—Rupture Disk in a Typical Holder 10
13—Combination of Reverse-Buckling Disk Assembly and Safety Relief Valve 11
14—Safety Relief Valve With O-Ring Seal 12
15—Acid Corrosion in Carbon-Steel Bonnet Caused by Leaking Seating Surfaces 14
16—Acid Corrosion on 18Cr-8Ni Steel Inlet Nozzle 14
17—Chloride Corrosion on 18Cr-8Ni Steel Nozzle (With Machined Seating Surface) 14
18—Sulfide Corrosion on Carbon-Steel Disk From Crude-Oil Distillation Unit 15
19—Chloride Attack on 18Cr-8Ni Steel Disk 15
20—Pit-Type Corrosion on 18Cr-8Ni Steel (Type 316) Bellows 15
21—Monel Rupture Disks Corroded in Sour-Gas Service 16
22—Body and Bonnet Coated With Epoxy for Corrosion Protection 16
23—Seating Surface of Disk Deformed by Chattering 16
24—Seat and Disk Damaged by Frequent Operation of Valve Set Too Close to Operating Pressure 17
25—Spring Failure Due to Corrosion 17
26—Spring Failure Due to Stress Corrosion 17
27—Inlet Nozzle Plugged With Coke and Catalyst After Nine Months in Reactor Vapor Line 18
28—Outlet of Valve Plugged With Deposits From Other Valves in Common Discharge Header 18
29—Moving Parts of Valve Fouled With Iron Sulfide (FeS_2) 19
30—Valve Stuck Because of Iron Sulfide (FeS_2) Deposits 19
31—Disk Frozen in Guide Because of Buildup of Products of Corrosion in Sour-Oil Vapor Service 20
32—Rough Handling of Valves Should Be Avoided 20
33—Identification Tag for Pressure-Relieving Device 22
34—Block Valves on Safety-Relief-Valve Inlet and Outlet Sealed Open and Bleeders Installed to Depressure Inlet and Outlet 25
35—Safety-Valve and Relief-Valve Leak Detector 27
36—Typical Safety-Valve and Relief-Valve Test Block Using Air as Test Medium 28
37—Sulfur Deposits in Body of Valve on Way to Repair Shop 29

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Inspection of Pressure-Relieving Devices

SECTION 1—GENERAL

1.1 Scope

This recommended practice describes automatic pressure-relieving devices commonly used in the oil and petrochemical industries. As a guide to the inspection and control of these devices in the user's plant, it is intended to ensure their proper performance. This publication covers such automatic devices as spring-loaded pressure relief valves, pilot-operated valves, and rupture disks.

The recommendations in this publication are not intended to supersede requirements established by regulatory bodies. This publication does not cover weak seams or sections in tanks, explosion doors, fusible plugs, control valves, and other devices that either depend on an external source of power for operation or are manually operated. Inspections and tests made at manufacturers' plants, which are usually covered by codes or purchase specifications, are not covered by this publication.

This publication does not cover training requirements for mechanics involved in the inspection and repair of pressure-relieving devices. Those seeking these requirements should see API 510, which gives the requirements for a quality control system and specifies that the repair organization maintain and document a training program ensuring that personnel are qualified.

1.2 Referenced Publications

The most recent editions of the following standards, codes, manuals, and specifications are cited in this recommended practice.

API

510 *Pressure Vessel Inspection Code: Inspection, Rating, Repair, and Alteration*

RP 520 *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries,* Part I, "Sizing and Selection," and Part II, "Installation"

RP 521 *Guide for Pressure-Relieving and Depressuring Systems*

Std 526 *Flanged Steel Safety-Relief Valves*

Std 527 *Seat Tightness of Pressure Relief Valves*

Std 620 *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*

Std 2000 *Venting Atmospheric and Low-Pressure Storage Tanks (Nonrefrigerated and Refrigerated)*

Bul 2521 *Use of Pressure-Vacuum Vent Valves for Atmospheric Pressure Tanks to Reduce Evaporation Loss*

ASME[1]

PTC 25.3 *Safety and Relief Valves*

Boiler and Pressure Vessel Code, Section I, "Power Boilers"; Section IV, "Heating Boilers"; Section VI, "Recommended Rules for Care and Operation of Heating Boilers"; Section VII, "Recommended Guidelines for Care of Power Boilers"; and Section VIII, "Pressure Vessels," Division 1

NACE[2]

Std MR0175 *Sulfide Stress Cracking Resistant-Metallic Materials for Oil Field Equipment*

NBBPVI[3]

NB-23 *National Board Inspection Code*

SECTION 2—PRESSURE-RELIEVING DEVICES

2.1 General

Pressure-relieving devices protect refinery equipment and personnel by automatically opening at predetermined pressures and preventing the destructive consequences of excessive pressures in process systems and storage vessels. Basic types include spring-loaded pressure relief valves, pilot-operated valves, rupture disks, weight-loaded devices, pressure-loaded devices, and pressure and/or vacuum vent valves.

A pressure-relieving device is actuated by inlet static pressure and designed to open during an emergency or under abnormal conditions to prevent a rise of internal fluid or vapor pressure to a value greater than a specified value. The device may also be designed to prevent excessive internal vacuum.

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[2]National Association of Corrosion Engineers, P.O. Box 218340, Houston, Texas 77218.
[3]National Board of Boiler and Pressure Vessel Inspectors, 1055 Crupper Avenue, Columbus, Ohio 43229.

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2.2 Types of Pressure–Relieving Devices

2.2.1 PRESSURE RELIEF VALVE

2.2.1.1 General

A pressure relief valve is an automatic pressure-relieving device that under excessive static upstream pressure opens and allows the product to flow until normal pressure has been restored. It is opened by a certain static pressure upstream of it to prevent the increase of the internal medium pressure to a value greater than a predetermined greatest safe value. Once the system pressure has decreased to a certain value less than the valve's set pressure, normal conditions have been restored and the valve automatically closes to prevent the further loss of product. The set pressure (opening pressure) and the closing pressure depend on the type and adjustment of the valve. It can be a safety valve, a relief valve, a safety relief valve, or a pilot-operated safety relief valve.

2.2.1.2 Safety Valve

2.2.1.2.1 General

A safety valve is a spring-loaded pressure-relieving device that is popped open by enough static pressure in the huddling chamber and kinetic energy of the gas or vapor to overcome the spring force on the disk as it is lifted, to provide full opening and allow minimal overpressure. The closing pressure will be less than the set pressure and will be reached after the blowdown phase is completed.

The spring of a safety valve is usually fully exposed outside of the valve bonnet to protect it from degradation due to the temperature of the relieving medium. The usual safety valve has a lifting lever for manual opening so that freedom of the working parts can be ensured. Open bonnet safety valves are not pressure tight on the downstream side. Figure 1 illustrates a full-nozzle, top-guided safety valve.

2.2.1.2.2 Applications

Safety valves are used on steam boiler drums and superheaters. They may also be used for general air and steam services in a refinery. Safety valve discharge piping may contain a vented drip pan elbow or a short piping stack vented to the atmosphere.

2.2.1.2.3 Limitations

Safety valves should not be used as follows:

a. In corrosive refinery services.
b. In any back pressure service (see Part I of API Recommended Practice 520).
c. Where the discharge must be piped to remote locations.
d. Where the escape of lading fluid around blowing valves is not desirable.
e. In liquid service.
f. As pressure control or bypass valves.

2.2.1.3 Relief Valve

2.2.1.3.1 General

A relief valve is a spring-loaded pressure-relieving device for liquid service that begins to open when the inlet pressure reaches set pressure, overcomes the spring force, and begins to lift the disk off the seat and that continues to open as inlet pressure and the lift of the disk increase, allowing the flow of the liquid to increase. The value of the closing pressure is less than that of the set pressure and will be reached after the blowdown phase is completed. Relief valve capacities are usually rated at either 10 percent or 25 percent overpressure, depending on the application.

These valves have closed bonnets to prevent the release of corrosive, toxic, or expensive fluids. They can be supplied with lifting levers, balancing bellows, and soft seats as needed. Figure 2 illustrates one type of relief valve. The ASME Code requires that liquid service relief valves installed after January 1, 1986 have their capacity certified and stamped.

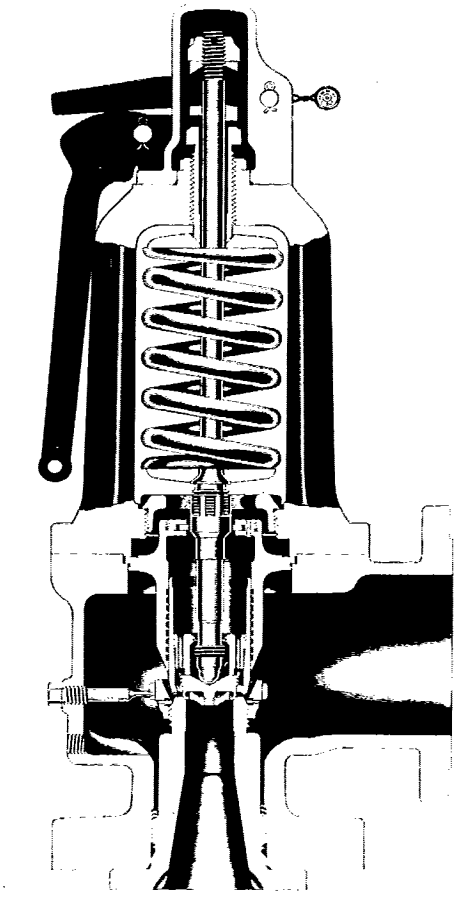


Figure 1—Top-Guided Safety Valve

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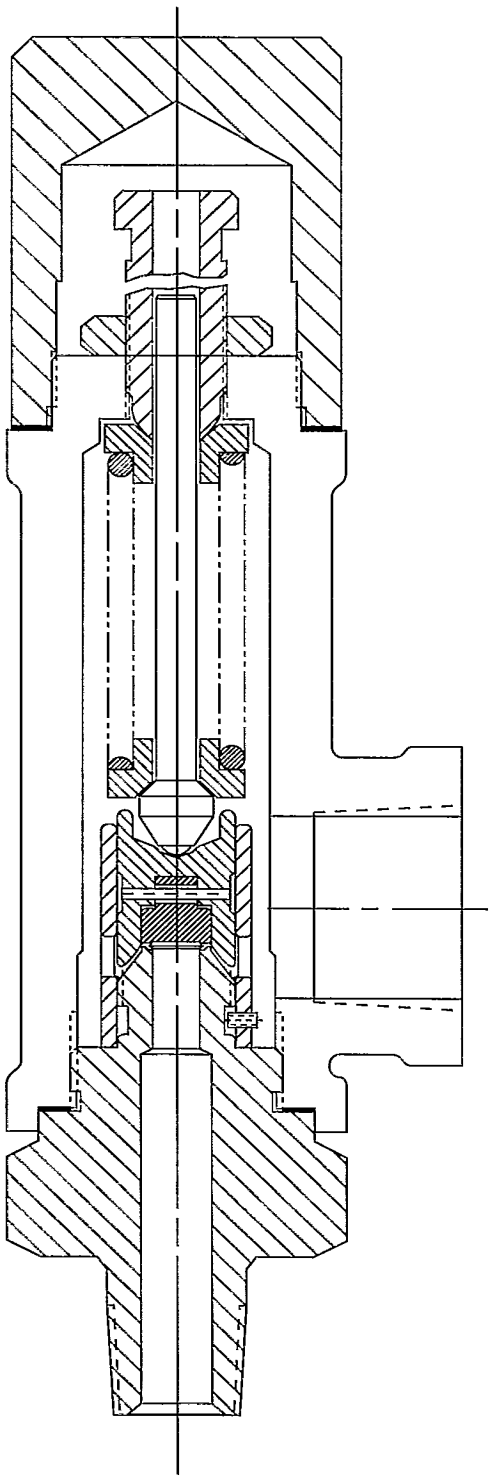


Figure 2—Relief Valve

2.2.1.3.2 Applications

Relief valves are normally used for incompressible fluids (see Part I of API Recommended Practice 520).

2.2.1.3.3 Limitations

Relief valves should not be used as follows:

a. In steam, air, gas, or vapor service.
b. In variable back pressure service unless the valves have a balancing bellows design or a piston design.
c. As pressure control or bypass valves.

The ASME Code requires that liquid service relief valves installed after January 1, 1986 have their capacity certified and stamped. Unstamped valves may not be used in liquid services.

2.2.1.4 Safety Relief Valve

2.2.1.4.1 General

A safety relief valve is a spring-loaded pressure relief valve that acts as a safety valve when used in gas or vapor service and as a relief valve when used in liquid service. A safety relief valve provided with a bonnet that encloses the spring and forms a pressure-tight housing is called *conventional*, and a safety relief valve that minimizes the effects of back pressure on its operational characteristics is called *balanced*. A safety relief valve opens fully before the overpressure reaches a value 10 percent greater than its set pressure in a compressible medium or 25 percent greater than its set pressure in an incompressible medium. See Figure 3.

2.2.1.4.2 Conventional Safety Relief Valve

2.2.1.4.2.1 General

A conventional safety relief valve has its spring housing vented to its discharge side. The operational characteristics (opening pressure, closing pressure, and relieving capacity) are directly affected by changes in the back pressure on the valve.

2.2.1.4.2.2 Applications

Conventional safety relief valves are used in refinery process industries that handle flammable, hot, or toxic material, which may pass through closed discharge systems and be released through the valves at safe, remote points. They are used as follow:

a. In general refinery service for gas, vapor, steam, air, or liquids.
b. In corrosive refinery service.
c. When the discharge from the valves must be piped to remote points.

See 2.2.1.4.2.3 for limitations on these uses.

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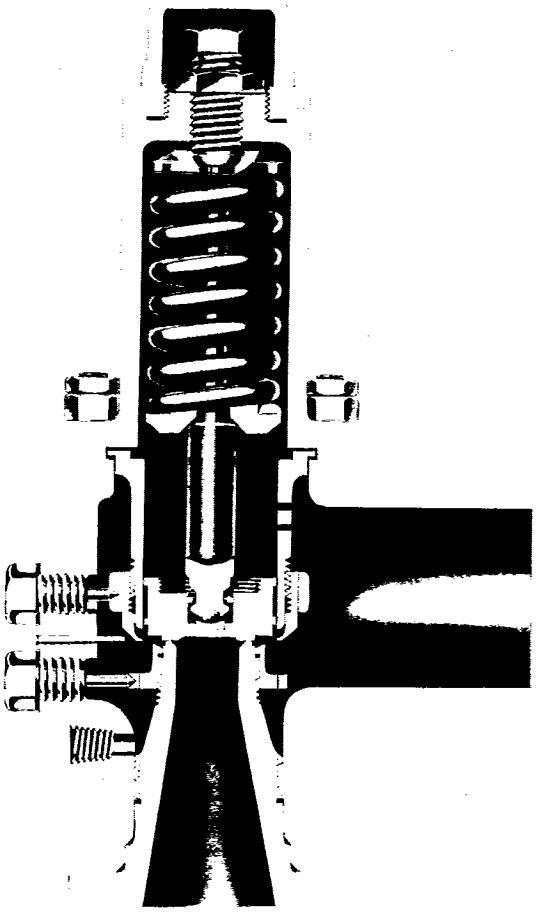


Figure 3—Conventional Safety Relief Valve

2.2.1.4.2.3 Limitations

Conventional safety relief valves may not be used in a service listed in 2.2.1.4.2.2 where any superimposed back pressure is constant or where any built-up back pressure exceeds 10 percent of the set pressure. Also, they should not be used as follows:

a. On steam boiler drums or superheaters.
b. In cases of variable back pressure (see Part I of API Recommended Practice 520).
c. As pressure control or bypass valves.

2.2.1.4.3 Balanced Safety Relief Valve

2.2.1.4.3.1 General

A balanced safety relief valve is a safety relief valve that incorporates a pressure-balancing bellows, piston, or both to minimize the effects of back pressure on its operational characteristics (opening pressure, closing pressure, and relieving capacity). Whether it is pressure tight on its downstream side depends on its design. A lifting lever is optional. A balanced bellows valve is shown in Figure 4. One with an auxiliary balancing piston is shown in Figure 5.

2.2.1.4.3.2 Applications

Balanced safety relief valves are normally used in refinery process industries that handle flammable, hot, or toxic material, which may pass through a closed discharge system and be released through the valve at a safe, remote point. They are used as follows:

a. In general refinery service for gas, vapor, steam, air, or liquids.
b. In corrosive refinery service.
c. When the discharge from the valves must be piped to remote points.

See 2.2.1.4.3.3 for limitations on these uses.

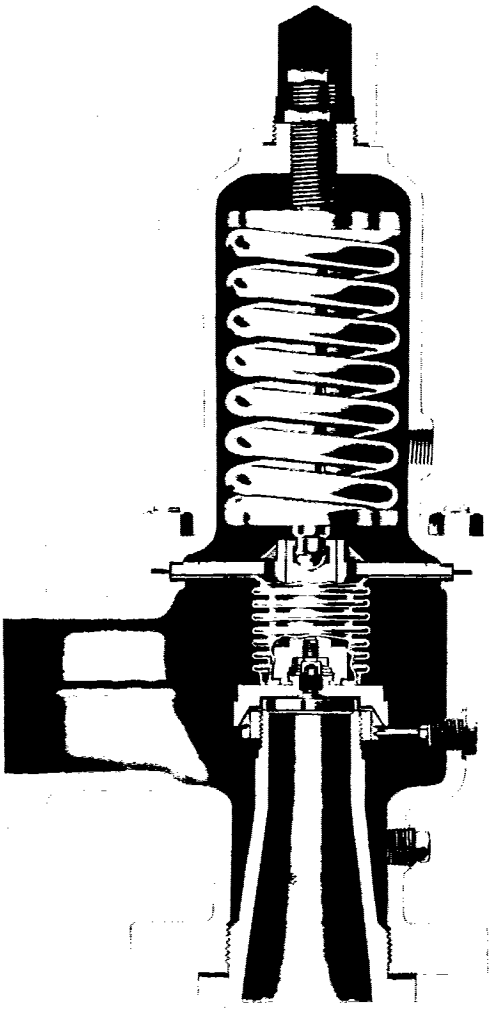


Figure 4—Balanced Bellows Safety Relief Valve

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2.2.1.4.3.3 Limitations

Balanced safety relief valves should not be used as follows:

a. On steam boiler drums or superheaters.
b. As pressure control or bypass valves.

Because balanced-type valves must have vented bonnets, the need to pipe the vent to a safe location must be considered. If the bellows fails, the balanced bellows valve, with or without an auxiliary balancing piston, will discharge fluid from the bonnet vent. The back pressure limitation listed in the valve manufacturer's catalog, which depends on the mechanical strength of the bellows, must not be exceeded.

2.2.1.5 Pilot–Operated Safety Relief Valve

2.2.1.5.1 General

A pilot-operated safety relief valve is a pressure relief valve whose main relieving valve is controlled by a small spring-loaded (set-actuated) pressure relief valve. Depending on the design, the pilot valve (control unit) and the main valve are mounted either on the same connection or separately. The pilot is a spring-loaded valve that measures the pressure and causes the main valve to open and close according to the pressure. Process pressure is either vented off by the pilot valve to open the main valve or applied to the top of the unbalanced piston, diaphragm, or bellows of the main valve to close it. Pilot-operated safety relief valves are required by Section VIII of the ASME Code to open and relieve at full capacity at not over the set pressure if some essential part of the pilot should fail. Figure 6 shows a low-pressure diaphragm-type pilot-operated valve. Each of Figures 7 and 8 shows a high-pressure pilot-operated valve that uses an unbalanced piston and has an integrally mounted pilot. Figure 8 also illustrates optional remote pressure sensing from the vessel and optional dual outlets to equalize thrust.

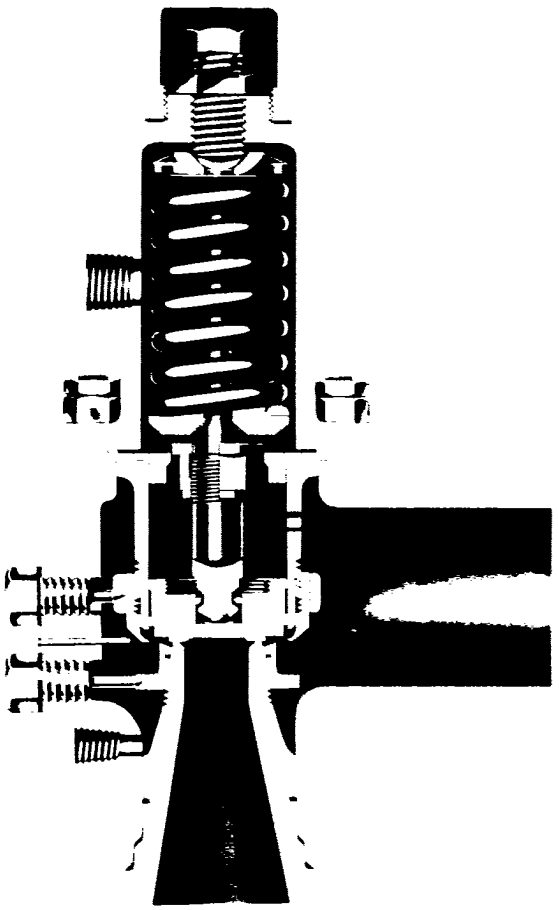


Figure 5—Balanced Bellows Safety Relief Valve With Auxiliary Balancing Piston

2.2.1.5.2 Applications

Pilot-operated safety relief valves are generally used as follows:

a. Where a large relief area at high set pressures is required, since many pilot-operated valves can be set to the full rating of the inlet flange.
b. Where a low differential exists between the normal vessel operating pressure and the set pressure of the valves.
c. On large low-pressure storage tanks (see API Standard 620).
d. Where very short blowdown is required.
e. Where back pressures are very high and balanced design is required, as pilot-operated valves with the pilots vented to the atmosphere are fully balanced.
f. In general process services where their use is economical and not limited by conditions in 2.2.1.5.3.
g. Where process conditions require sensing of pressure at one location and relief of fluid at another location.

2.2.1.5.3 Limitations

Pilot-operated safety relief valves are not generally used as follows:

a. In service where fluid is dirty.
b. In viscous liquid service, as pilot-operated valves have small orifices that become plugged by viscous liquids.
c. With vapors that will polymerize in the valves.
d. In service where the temperature exceeds the safe limits for the diaphragms or seals selected.
e. Where chemical compatibility of the lading fluid with the diaphragms or seals of the valves is questionable or where corrosion buildup can impede the actuation of the pilots.

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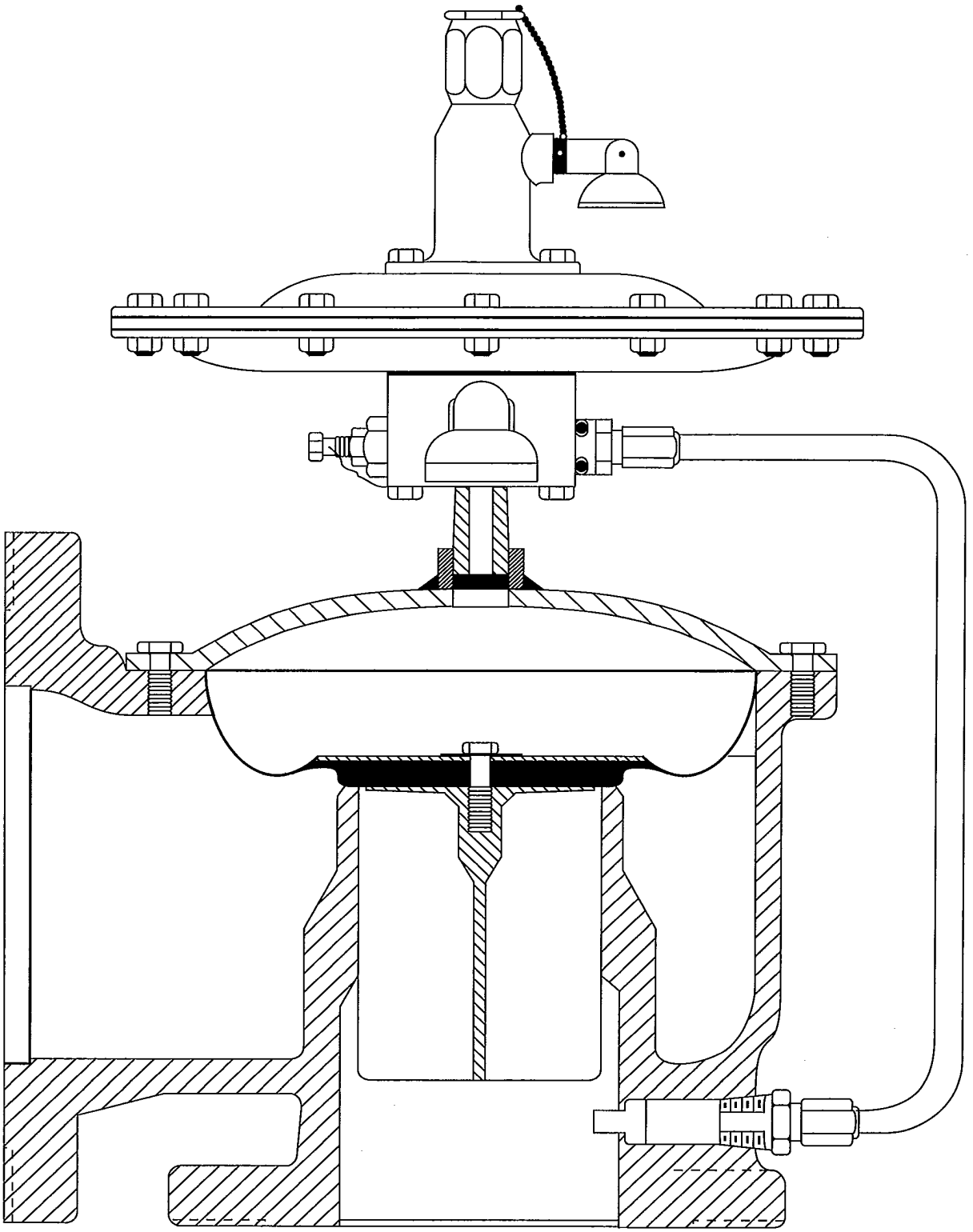


Figure 6—Low-Pressure Diaphragm-Type Pilot-Operated Valve

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2.2.2 PRESSURE AND/OR VACUUM VENT VALVE

2.2.2.1 General

A pressure and/or vacuum vent valve (also known as *a pressure and/or vacuum relief valve*) is an automatic pressure- or vacuum-relieving device actuated by the pressure or vacuum in the protected storage tank. A pressure and/or vacuum vent valve falls into one of three basic categories:

a. Weight-loaded pallet vent valve, as shown in Figure 9.
b. Pilot-operated vent valve, as shown in Figure 6.
c. Spring- and weight-loaded vent valve, as shown in Figure 10.

2.2.2.2 Applications

Pressure and/or vacuum vent valves are used to protect atmospheric and low-pressure storage tanks against a differential pressure—the difference between the pressure or vacuum inside the tank and the atmospheric pressure outside the tank—large enough to damage the tank. Breather valves, single units composed of both pressure vent valves and vacuum vent valves and also known as *conservation vent valves*, are suggested for atmospheric storage tanks containing oil with a flash point below 100°F (38°C). However, they may also be used on heavier oils (see API Bulletin 2521 and API Standard 2000).

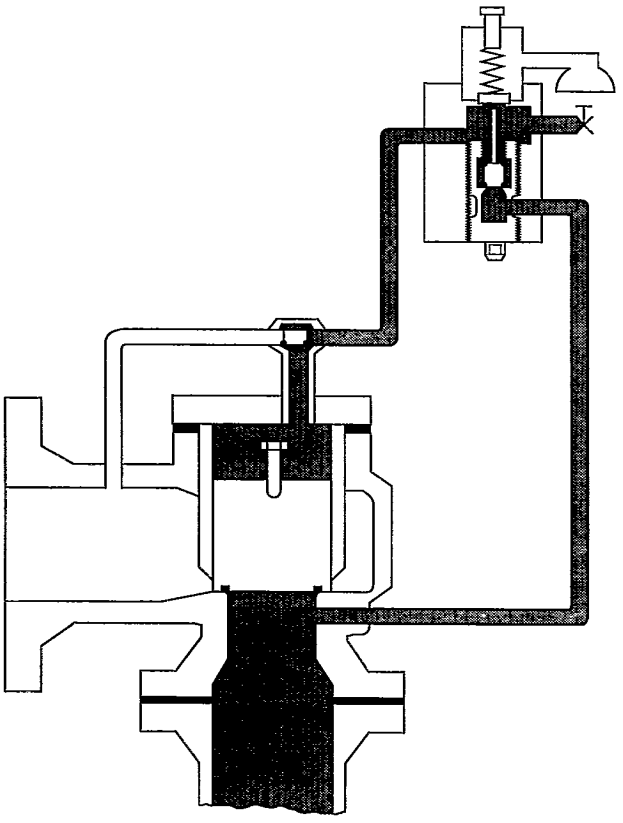


Figure 7—High-Pressure Pilot-Operated Valve

2.2.2.3 Limitations

Pressure and/or vacuum vent valves are designed to protect atmospheric storage tanks and should generally not be used otherwise.

2.2.3 RUPTURE DISK DEVICE

2.2.3.1 General

A rupture disk is a thin diaphragm normally held between special flanges that relieves pressure from a vessel or system by rupturing at a predetermined pressure. Figure 11 shows a typical preformed, or pre-bulged, rupture disk. Rupture disks are available in various metals, plymetals, plastic-metals, and coated metals. Nonmetallic disks, such as those of impervious graphite, are flat. The combination of a rupture disk and its holder is known as *a rupture disk device* (or *a nonreclosing pressure-relieving device*). A typical rupture disk device is illustrated in Figure 12.

Some bulged metal disks are designed for installation with the concave side toward the source of pressure. This places the disk material in tension. If the protected system operates with continuous or periodic vacuum, a vacuum support may be required.

Shown combined with a safety relief valve in Figure 13, the reverse-buckling rupture disk is normally good for vapor service only and should not be used in full liquid service without consulting the manufacturer. The preformed reverse-buckling disk is installed with the convex side of the bulge facing the pressure source, so that the disk material is in compression and no vacuum support is required. If the system pressure rises to the predetermined level, the disk will suddenly buckle or completely reverse in a snap. The rupture disk opens after reversal either by tearing along score lines or by being cut or sheared by knife blades or teeth, thus providing a full-flow opening.

An impervious graphite rupture disk has several unique characteristics. Its advantages include high corrosion resistance, resistance to fatigue, immunity to thermal shock within its design range, and bursting accuracy independent of temperature within its design temperature range. However, it is flat and requires vacuum support when vacuum is expected in the protected vessel or system.

When a graphite rupture disk is used on a vessel or system or with a relief valve, related piping should be provided with traps to prevent pieces of graphite from falling back into the system and to protect the upstream side of the relief valve.

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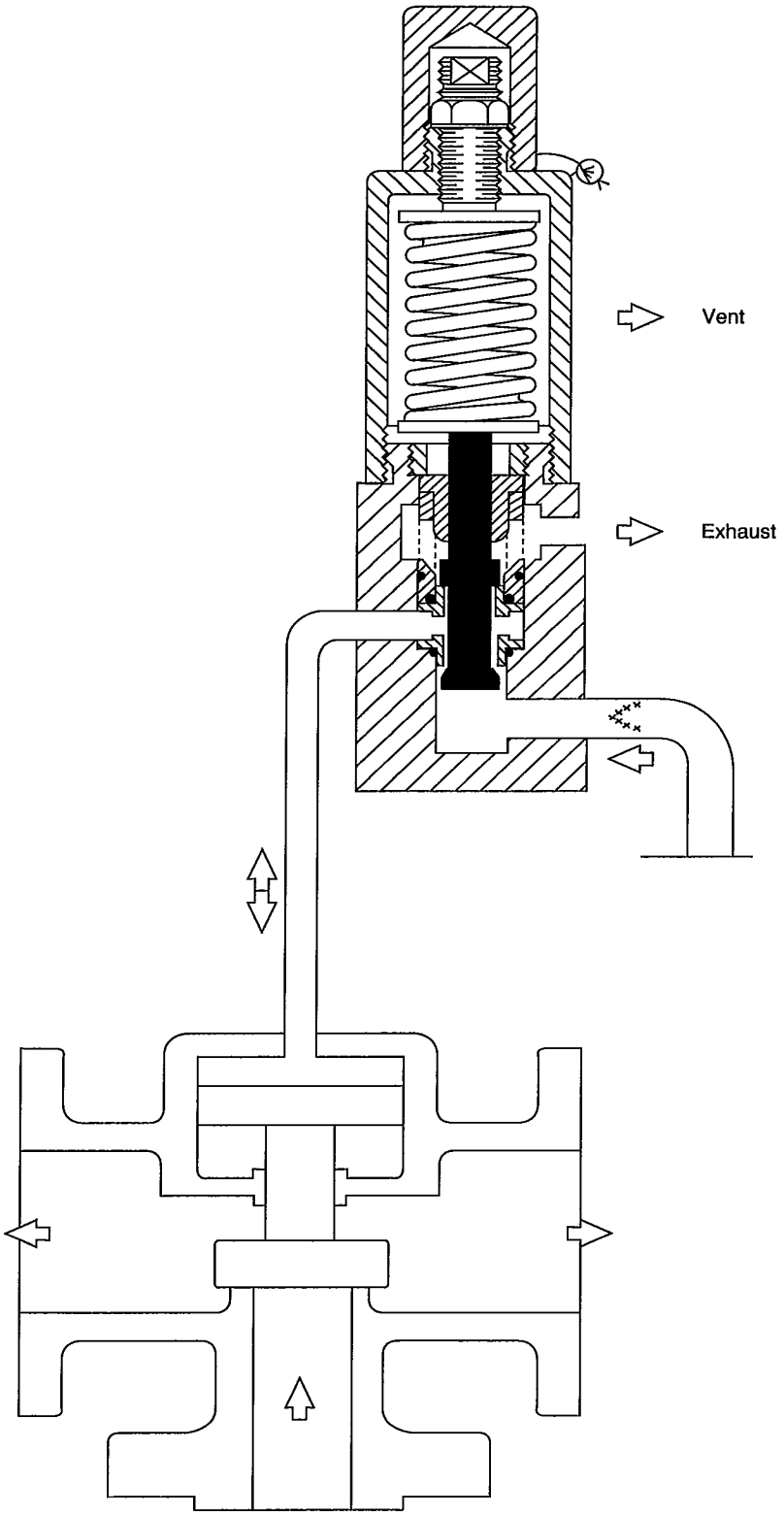


Figure 8—High-Pressure Pilot-Operated Valve With Optional Dual Outlets

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2.2.3.2 Applications

Rupture disk devices are sometimes used as follows:

a. To protect the upstream side of pressure relief valves against corrosion by the system fluid.
b. To protect pressure relief valves against plugging or clogging by viscous fluids or polymerization products.
c. Instead of pressure relief valves if the protected system can tolerate process interruptions or loss of fluids in case the disks rupture.
d. As secondary pressure-relieving devices when the difference between the operating pressure and the rupture pressure is large, depending on the type of rupture disk selected (see 2.2.3.1 and 2.2.3.3).
e. To protect the downstream sides of pressure relief valves against downstream corrosion from headers or against atmospheric corrosion.
f. To minimize leakage.

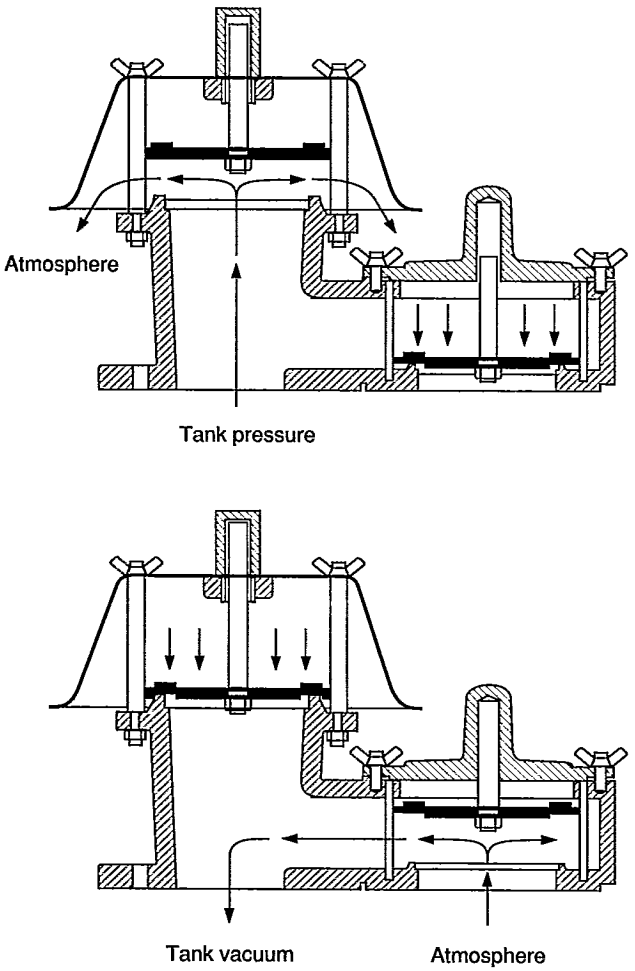


Figure 9—Weight-Loaded Pallet Vent Valve

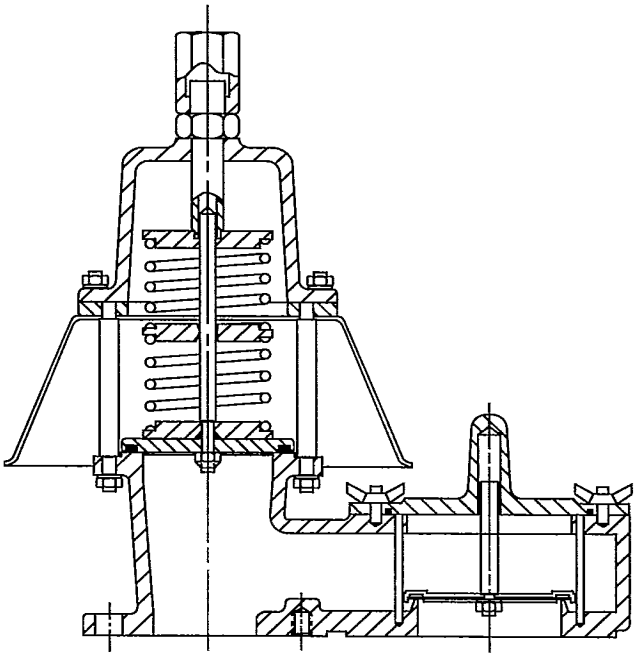


Figure 10—Spring- and Weight-Loaded Vent Valve

The proper receipt, storage, handling, and installation of a rupture disk are critical to its successful performance. The manufacturer's instructions, especially those concerning limits on bolt torque, should be referred to and followed.

When installing rupture disks upstream of safety relief valves, several important recommendations in Section VIII of the ASME Code should be considered. For example, either the capacity of the safety relief valve should be limited to 90 percent of its rated relieving capacity or the capacity of the combination should be established in accordance with ASME Code rules (see 2.2.3.3). When rupture disk installations are contemplated in general, the ASME Code and any other applicable state or local codes and company standards should be consulted.

When knife blades have been used to open rupture disks, the disks have often not opened properly. The manufacturer should be consulted concerning the proper installation of rupture disks to be opened with knife blades.

2.2.3.3 Limitations

For bulged metal rupture disks installed so that pressure acts against the concave side and for flat metal rupture disks, the operating pressure of the protected system is usually limited to 65–85 percent of the disk's predetermined bursting pressure. The exact percentage depends on the type of disk used. The lower value of this range should be used if either of the following is anticipated in the system:

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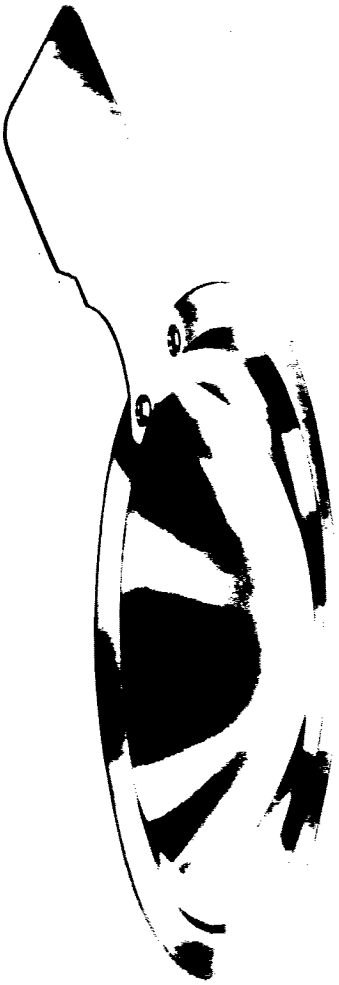


Figure 11—Rupture Disk

a. Pulsating pressure.
b. Continuous or periodic wide swings in pressure.

The service life of these disks under normal operating conditions is usually one year. They are subject to relatively rapid creep stress failure, especially in high operating temperatures. If not replaced periodically, they may fail without warning at normal operating pressures.

Higher operating pressures—up to 90 percent of the disk design bursting pressure—are possible with the reverse-buckling disk. Less fatigue due to pulsating and cyclic operating pressure results in a longer service life than would be expected if the disk were installed with the pressure acting against the concave side. Most reverse-buckling disks should not be used in full liquid service. However, if an assured gas pocket rests against the disk and the disk manufacturer is consulted, liquid service can be considered. With finite lives, these disks should be replaced periodically. Consult the manufacturer for recommended replacement times.

Impervious graphite rupture disks offer nearly the same advantages and disadvantages as the reverse-buckling, metal type. However, with impervious graphite rupture disks, the piping arrangement may be more complicated (see 2.2.3.1), and uneven flange bolt loads or thermal strains in the piping may crack the disk.

CAUTION: When rupture disks are removed for inspection or when an accompanying relief valve is serviced, the disks could easily be damaged and could even fail prematurely if reused. Replacement of disks at every maintenance interval will minimize the chance of damage and failure.

When designing or setting up an installation using rupture disks, the disk manufacturer should be consulted about the probable bursting accuracy of their disks when used in various sizes, designs, temperatures, and service pressures.

A pressure gauge, a try cock, a free vent, or a suitable tell-tale indicator should be inserted between a rupture disk device installed at the inlet of a pressure relief valve

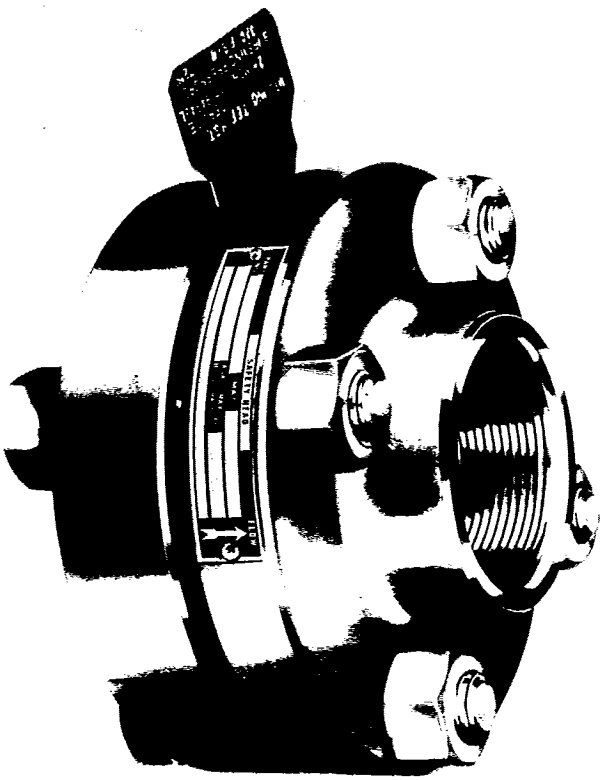


Figure 12—Rupture Disk Device—Rupture Disk in a Typical Holder

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and the valve (see Figure 13), permitting the detection of disk rupture or leakage. A rupture disk device must have full pipe area and must not fragment into the safety relief valve inlet after the disk bursts. When a rupture disk device is used with a pressure relief valve, ASME Code rules on the capacity of the rupture disk must be followed (see 2.2.3.2).

2.3 Variations With Resilient Valve Seats

2.3.1 GENERAL

Some relief valves and safety relief valves are manufactured with resilient O rings or other types of soft seals to supplement or replace the conventional metal-to-metal valve seating surfaces. Usually the valves are similar in all respects to the basic valves described above, with the exception that the disks are designed to accommodate some type of seal ring to promote a degree of tightness exceeding that of the usual commercial tightness of conventional metal seats. Figure 14 shows one type of O-ring seat seal.

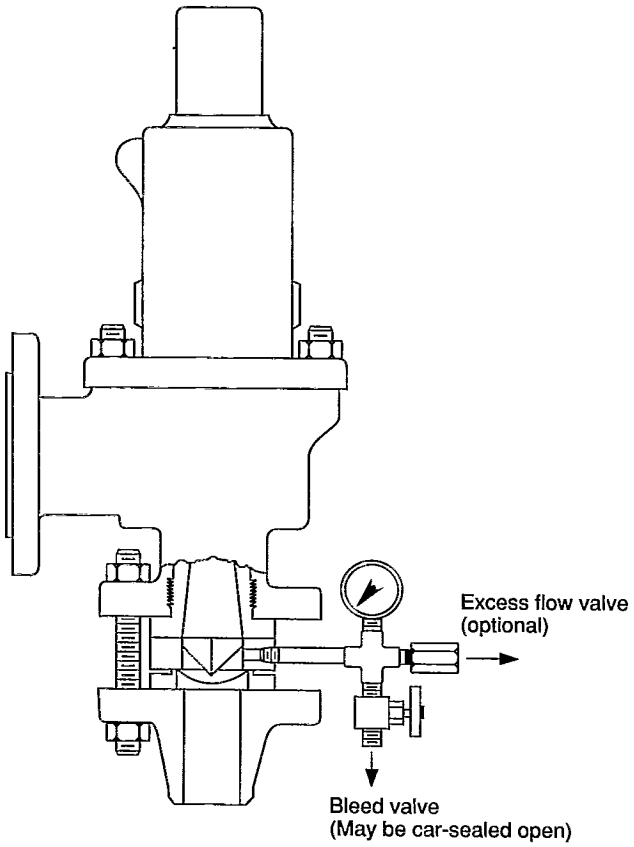


Figure 13—Combination of Reverse-Buckling Disk Assembly and Safety Relief Valve

2.3.2 APPLICATIONS

Resilient valve seats are frequently used when a greater degree of seat tightness is required than is likely to occur with metal-to-metal seats. Resilient valve seats are also used as follows:

a. *Where the service fluid is corrosive or hard to hold with metal-to-metal seats.* Slight leakage of a corrosive gas, vapor, or liquid could deteriorate or foul the moving parts of a valve with corrosion products.

b. *To prevent the loss or to minimize the escape of explosive, toxic, or irritant fluids or of expensive products.*

c. *Where operating pressures may have to be very close to the set pressure.* As the operating pressure approaches the valve set pressure, the net differential forces on the disk are reduced. This decreases seat tightness. Resilient valve seats provide greater tightness than do metal-to-metal types as the operating pressure approaches the valve set pressure.

d. *When small, hard, foreign particles are carried in flowing fluids.* In this case, the particles usually scratch or mar metal-to-metal seats when the valve discharges, usually resulting in leakage when the valve closes. The resilient seal can absorb the impacts of the particles, shield the mating metal seating surface, and reduce the likelihood of leakage.

e. *When a safety relief valve is subject to a minor pressure surge.* In this case, the disk may only lift enough to cause a slightly audible escape of fluid or a barely visible drip of liquid. Although the system pressure is relieved, the valve does not pop or lift wide open. In this case, with metal-to-metal seats, the disk may not reseat properly and the valve may continue to leak until the system's pressure is below its normal operating pressure. A resilient seat valve provides tight shutoff when the system pressure falls after minor pressure relief.

f. *Where safety valves and relief valves are subject to vibration and pulsating pressure.* The vibration and pulsating pressure tend to reduce the effect of the spring load on the disk. The reduced spring load effect causes the disk to rub against its seat, resulting in leakage. In this case, resilient seals can maintain a greater degree of tightness than metal-to-metal seats.

2.3.3 LIMITATIONS

A variety of resilient materials is available for seals in valves. No single material is yet suitably resistant to all pressures, temperatures, chemicals, and other influences. Certain materials may be excellent with respect to chemical resistance, but unsuitable for the intended service temperatures, and vice versa. Therefore, the specific service conditions should be considered in selecting a specific resilient seat arrangement.

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The use of O-ring seal rings is limited to a practical pressure range. Extremely high pressure may cause extrusion of the O ring from its retaining groove or damage to the ring and loss of its proper functioning.

Past plant experience is the best guide in choosing seal material and pressure limitations. In case this experience is unavailable, the valve manufacturers and their catalogs may be consulted.

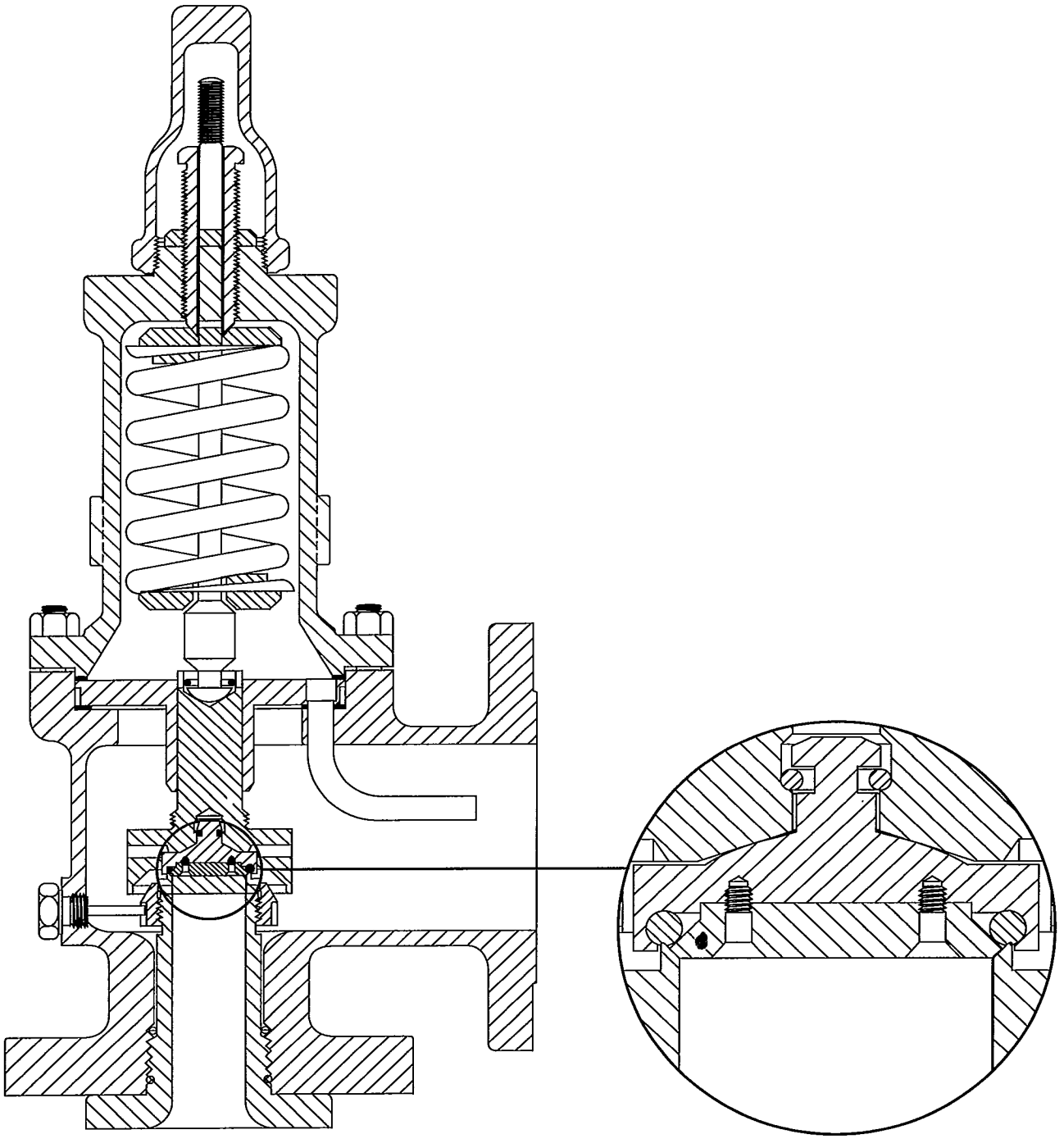


Figure 14—Safety Relief Valve With O-Ring Seal

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SECTION 3—REASONS FOR INSPECTION

3.1 To Protect Personnel and Equipment

Pressure-relieving devices are installed on process equipment to release excess pressure due to upset operations, external fires, and other hazards. These hazards are included in Table 2 of Part I of API Recommended Practice 520 and discussed in API Recommended Practice 521. Failure of pressure-relieving devices to function properly when needed could result in the overpressure of the vessels, exchangers, boilers, or other equipment they were installed to protect. A properly designed, applied, and installed pressure-relieving device that is maintained in good operating condition is essential to the safety of personnel and the protection of equipment during abnormal circumstances. One of the principal reasons for inspecting pressure-relieving devices is to ensure that they will provide this protection.

3.2 To Determine the Condition of a Device

Inspections of pressure-relieving devices must determine the general physical and operating conditions of the devices, to ensure that their performances meet the requirements for a given installation. In making this determination, two areas of inspection are involved:

a. *Shop inspection of the pressure-relieving devices* to determine their internal conditions, operability on the test stand, and conformance to specifications (see 7.2.1).
b. *Inspection of the protected equipment* to find any conditions or situations that might impede the proper functioning of the pressure-relieving devices, including the deposit of coke or corrosion that could foul moving parts or restrict inlet or outlet flow. Inlet and outlet piping should be inspected for the presence of internal deposits, and records should be kept of their condition. If deposits are observed, the piping should be thoroughly cleaned. If necessary, piping should be dismantled for inspection and cleaning. Visual on-stream inspection should be made periodically (see 5.2.6).

3.3 To Evaluate Inspection Frequency

The inspection of pressure-relieving devices provides data that can be evaluated to determine a safe and economical frequency of scheduled inspections. This frequency varies widely with the various operating conditions and environments to which relief devices are subjected. Inspections may usually be less frequent when operation is satisfactory and more frequent when corrosion, fouling, and leakage problems occur. Historical records reflecting periodic test results and service experiences for each relief device are valuable guides for establishing safe and economical inspection frequencies.

SECTION 4—CAUSES OF IMPROPER PERFORMANCE

4.1 Corrosion

Nearly all types of corrosion are present in refinery service, and corrosion is the basic cause of most difficulties encountered. Corrosion is often apparent: pitted or broken valve parts, deposits of corrosive residue that interfere with the operation of the moving parts, or a general deterioration of the material of the relieving device. Figures 15 through 21 illustrate the effects of corrosion on relieving devices. In addition to internal parts, exposed studs are vulnerable to environmental corrosion attacks.

Corrosion can usually be slowed or stopped by the selection of more suitable devices or device materials or by better workmanship in the shop's maintenance and repair section to ensure greater valve tightness. A leaking valve allows corrosive fluids to circulate in the upper parts of the valve, which can contribute to the corrosion of its movable parts. The corrosion protection coatings shown in Figure 22 are applicable in some services.

In many instances, valves of different construction can avoid, reduce, or even completely contain the effects of the corrosion. The use of an O-ring seat seal in a safety relief valve will sometimes stop leakage past the seating surface and eliminate corrosion in the valve's working parts (see Figure 14). If it is impossible to eliminate leakage entirely or if corrosive back pressure is exerted, a bellows seal can be used to protect the moving parts of the valve from the corrosive fluid (see Figure 4).

4.2 Damaged Seating Surfaces

Because differential loading must be small to prevent leakage of the lading fluid, an optical precision on the order of 3 light beads/bands [0.0000348 inch (0.0008838 mm)] must be maintained in the flatness of seating surfaces on pressure relief valves (see API Standard 527). Any imperfection in these seating surfaces will contribute to improper valve action in service.

There are many causes of damaged valve seats in refinery or chemical plant service, including the following:

a. *Corrosion.*
b. *Foreign particles that get into the valve inlet and pass through the valve when it opens,* such as mill scale, welding

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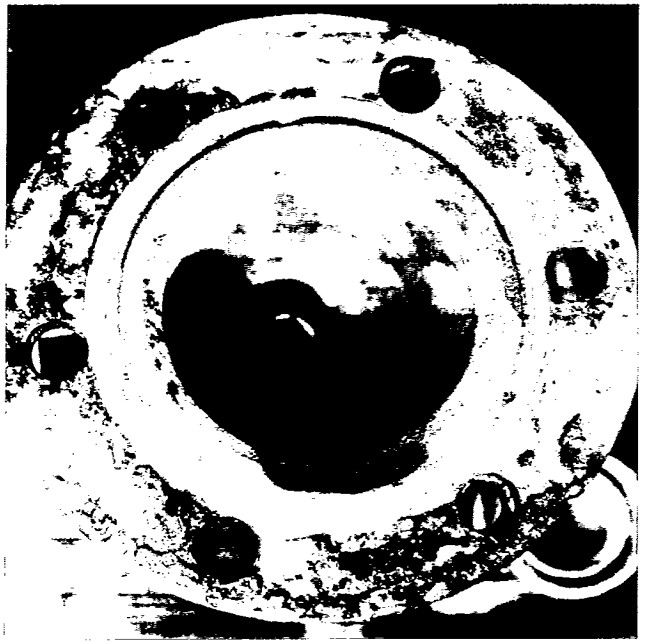


Figure 15—Acid Corrosion in Carbon-Steel Bonnet Caused by Leaking Seating Surfaces

spatter or slag, corrosive deposits, coke, or dirt. The particles may destroy the fineness of seat contact required for tightness in most pressure relief valves. The destruction can occur either in the shop during maintenance of the valve or in the protected unit while the valve is in use.



Figure 16—Acid Corrosion on 18Cr-8Ni Steel Inlet Nozzle

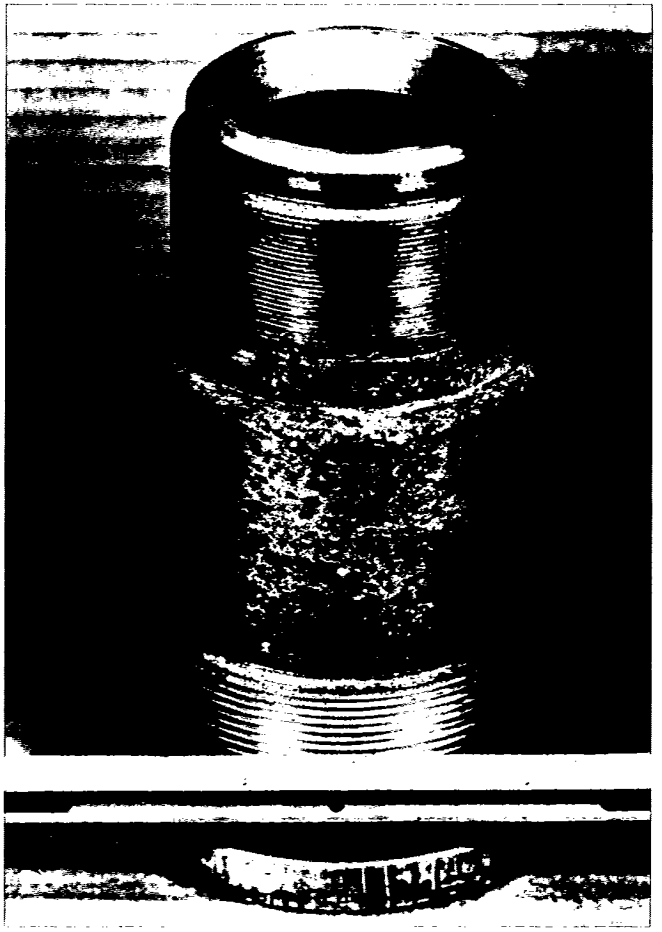


Figure 17—Chloride Corrosion on 18Cr-8Ni Steel Nozzle (With Machined Seating Surface)

c. *Improper or lengthy piping to the valve inlet or obstructions in the line.* These cause the valves to chatter (hammer). The pressure under the seat may become great enough to open the valve. However, as soon as the flow is established, the pressure drop in the connecting piping may be so great that the pressure under the seat falls and allows the valve to close. A cycle of opening and closing may develop, become rapid, and subject the valve seating surfaces to severe hammering, which damages the seating surfaces, sometimes beyond repair. Figures 23 and 24 show seating surfaces damaged by chattering and frequent fluctuations of pressure.

d. *Careless handling during maintenance,* such as bumping, dropping, jarring, or scratching of the valve parts.

e. *Leakage past the seating surfaces of a valve after it has been installed.* This leakage contributes to seat damage by causing erosion (wire drawing) or corrosion of the seating surface and thus aggravating itself. It may be due to improper maintenance or installation such as misalignment of the parts, to piping strains resulting from improper support, or to complete lack of support of discharge piping.

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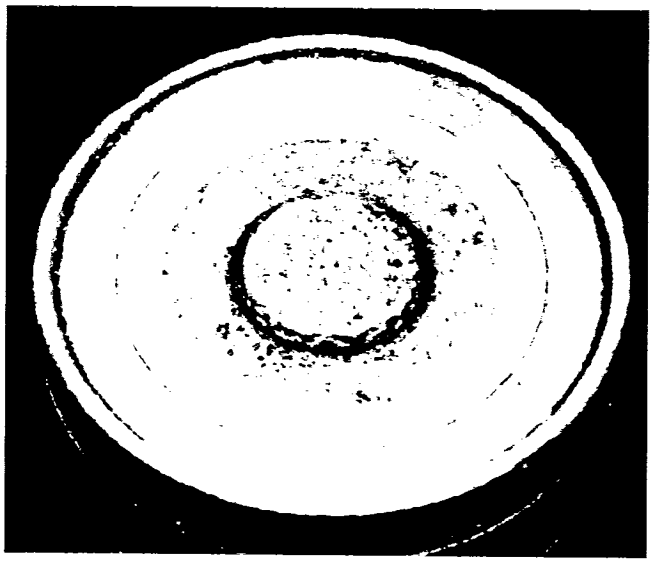


Figure 18—Sulfide Corrosion on Carbon-Steel Disk From Crude-Oil Distillation Unit

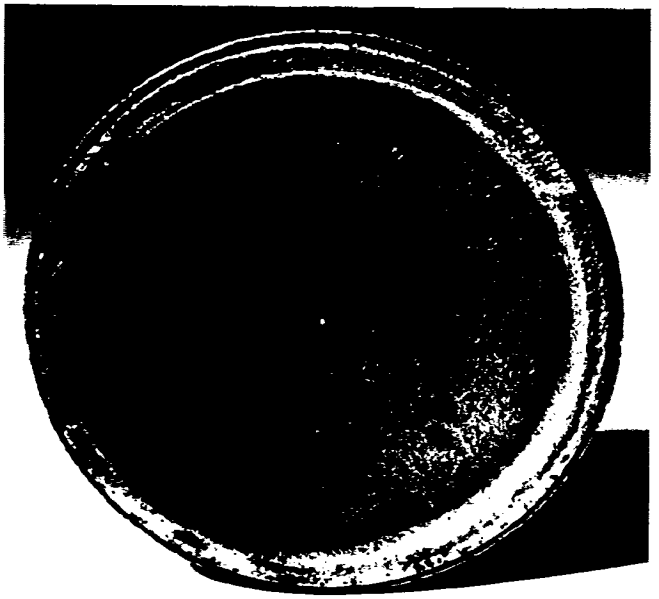


Figure 19—Chloride Attack on 18Cr-8Ni Steel Disk

Other frequent causes of this leakage are improper alignment of the spindle, improper fitting of the springs to the spring washers, and improper bearing between the spring washers and their respective bearing contacts or between the spindle and disk or disk holder. Spindles should be checked visually for straightness. Springs and spring washers should be kept together as a spring assembly during the life of the spring.

f. *Improper blowdown ring settings.* These can cause chattering in pressure relief valves. The relief valve manufacturer should be contacted for specific blowdown ring settings for liquid service and for vapor service.

4.3 Failed Springs

Spring failure occurs in two forms. The first is a weakening of the spring, which causes a reduction in set pressure and the possibility of premature opening. The second is a total failure (complete break) of the spring, which causes uncontrolled valve opening.

Failed springs in pressure relief valves are almost always caused by corrosion. Two types of corrosion are most prevalent in this type of failure in refineries:

a. *Surface corrosion* attacks the spring surface until the cross-sectional area of the spring is not sufficient to provide the necessary strength. It may also produce pits that act as stress risers and cause cracks in the spring surface and subsequent spring failure (see Figure 25).

b. *Stress corrosion* sometimes causes rapid spring failure. It is especially insidious because it is very difficult to detect before the spring breaks. A brittle-type spring failure due to stress corrosion is shown in Figure 26. Hydrogen sulfide

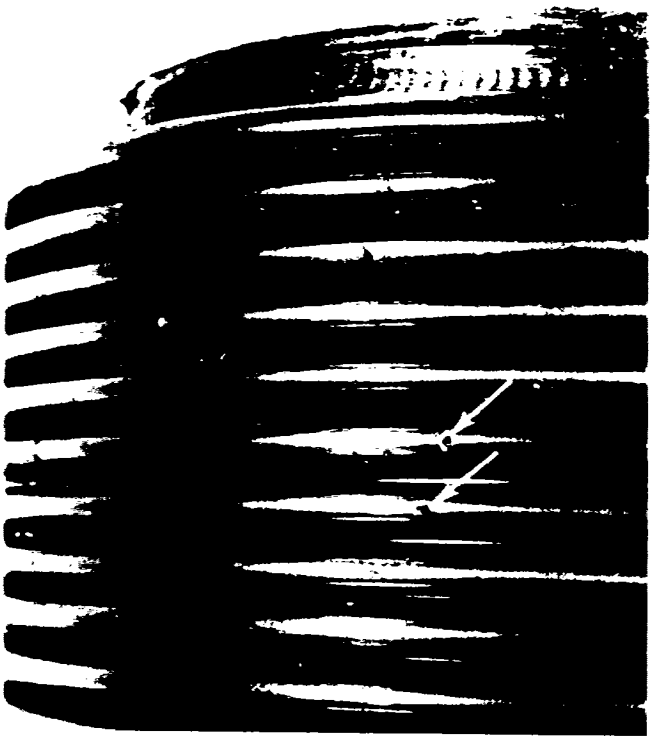


Figure 20—Pit-Type Corrosion on 18Cr-8Ni Steel (Type 316) Bellows

(H_2S) frequently causes stress-corrosion spring failure in petrochemical plants (see NACE Standard MR0175 for material recommendations and guidance).

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Figure 21—Monel Rupture Disks Corroded in Sour-Gas Service

Whether a spring fails at any given time depends on the type and strength of the corrosive, the stress level in the spring, the spring material, and how long the spring has been in use.

Where corrosion prevails, three courses of preventive action may be taken:

a. A spring material that will satisfactorily resist the action of the corrosive agent may be used.
b. The spring may be isolated by a bellows.
c. The spring may be specially coated to prevent corrosion by a coating that can stand the operating temperature and environment.

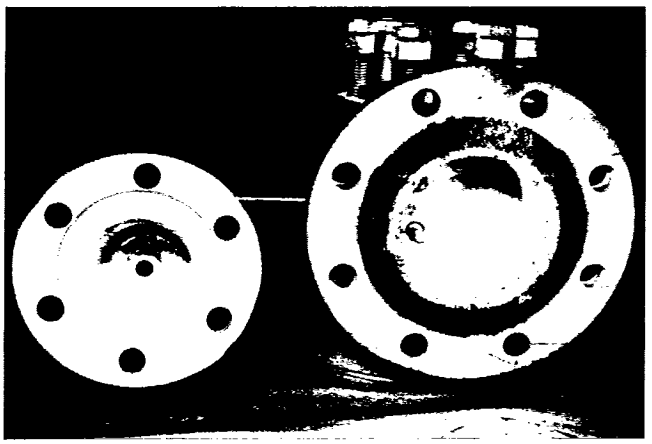


Figure 22—Body and Bonnet Coated With Epoxy for Corrosion Protection



Figure 23—Seating Surface of Disk Deformed by Chattering

4.4 Improper Setting and Adjustment

Improper setting of a pressure relief valve is usually due to carelessness by or inadequate training of maintenance personnel, reliance on inadequate testing equipment in the maintenance shop, or inadequate knowledge of how to properly adjust the pressure relief valve. Manuals provided by valve manufacturers help eliminate those deficiencies by indicating how to adjust their valves for temperature, back pressure, and other factors.

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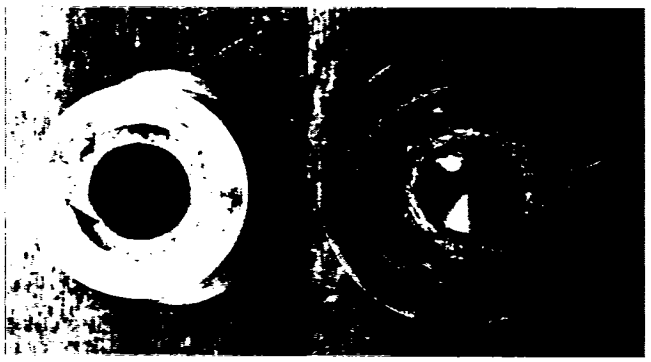


Figure 24—Seat and Disk Damaged by Frequent Operation of Valve Set Too Close to Operating Pressure

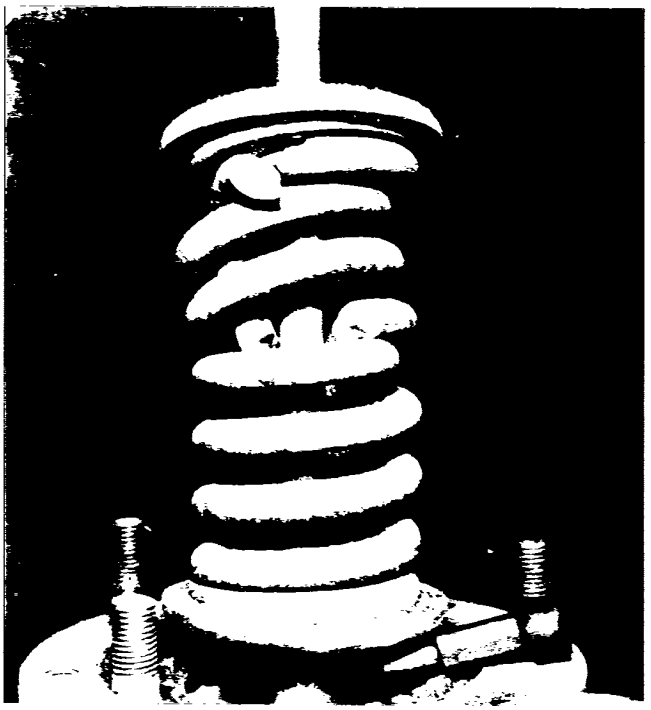


Figure 25—Spring Failure Due to Corrosion

In refinery and chemical plant service, setting a pressure relief valve while it is in place on the equipment to be protected and by using the fluid that will flow through it during operation is usually impractical. Therefore, valves other than boiler safety valves must be set in the valve maintenance shop while on appropriate test equipment. During inspection and repair, a properly designed test block facilitates the setting and adjusting of the pressure relief valve (see 6.2). For an explanation of nameplate terms required by repair work, see API Standard 526. Also see Section VIII, Division 1, paragraph UG-129, of the ASME Code for instructions on marking nameplates of pressure-relieving devices.

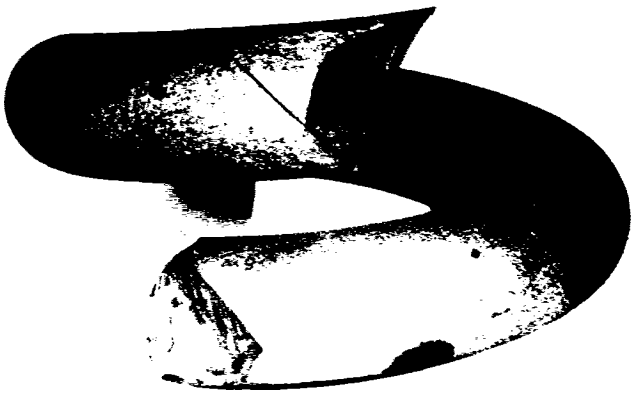


Figure 26—Spring Failure Due to Stress Corrosion

Water, air, or an inert gas such as bottled nitrogen is generally used as the testing medium in the shop. It is better to set a safety relief valve on air or some other gas than on water since, depending on the type of valve being tested, a gaseous effluent will produce either a definite pop or a clearly defined audible opening at the set pressure. To ensure that the valve is opening, some overpressure should be carefully applied because an audible leak could otherwise be misinterpreted as the result of reaching the set pressure. However, most safety relief valves produce a distinct pop at the set pressure, making misinterpretation impossible. As a test stand without sufficient surge volume might *not* cause a distinct pop, a test stand of the wrong size might cause an faulty set pressure. Vapor service valves should be set using air. Steam service valves should be set using steam, but air may be used if suitable corrections are applied. Liquid service valves should be set using water.

Incorrect calibration of pressure gauges is another frequent cause of improper valve setting. To ensure accuracy, gauges should be calibrated frequently on a regularly calibrated dead weight tester. The pressure range of the gauge should be chosen so that the required set pressure of the pressure relief valve falls within the middle third of the gauge pressure range. Snubbers on pressure gauges are not generally recommended since they tend to clog and produce pressure lag.

How to adjust the ring or rings controlling the valve is frequently misunderstood. The valve adjusting ring or rings will control either the valve blowdown—the difference between the set pressure and the reseating pressure—or valve blowdown and simmer, depending on the design of the valve being tested. Because the density and expansion characteristics of material handled through pressure relief valves are variable and the size of testing facilities is limited, it is usually impossible to adjust the rings on a test drum in the maintenance shop for the best possible performance. The rings should therefore be adjusted to obtain a pop on the valve test drum and then inspected and re-

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adjusted for proper blowdown according to the manufacturer's recommendation. This should permit the best average performance characteristics of the valve when installed. For liquid or vapor service, the relief valve manufacturer should be contacted regarding the proper blowdown ring settings. Full understanding of terminology is important (see ASME PTC 25.3).

4.5 Plugging and Sticking

In petrochemical service, process solids such as coke or solidified products can sometimes plug various parts of the valve and connected piping. Extreme cases of fouling are illustrated by Figure 27, which shows the inlet nozzle of a valve plugged solid by a mixture of coke and catalyst, and Figure 28, which shows the outlet nozzle of a valve plugged with deposits from other valves that discharge into a common discharge header. Fouling of a lesser degree, as shown in Figures 29 and 30, is also likely to impede valve operation. All valve parts, particularly guiding surfaces, should be checked thoroughly for any type of fouling. Lubrication of all sliding surfaces and threads with molybdenum disulfide (MoS_2) compounds or graphite and grease is recommended.

Valve malfunction may also be due to sticking of the disk or disk holder in the guide, which may be caused by corrosion or galling of the metal or by foreign particles in the guiding surfaces. Figure 31 shows a disk that is frozen in the guide as a result of corrosion in sour gas service. If corrosion is the cause of the sticking, two possible cures are available. First, the use of a bellows can protect moving parts from the corrosive substance, especially in closed systems (see Figure 4). Second, an O-ring seal (see Figure 14) can seal the guiding surfaces from the lading fluid. Foreign particles in the guiding surfaces tend to roll metal up, causing severe galling. The use of a bellows can keep the foreign particles away from the guiding surfaces. Sticking of valves can also result from machining of valve parts outside the manufacturer's tolerance limits.

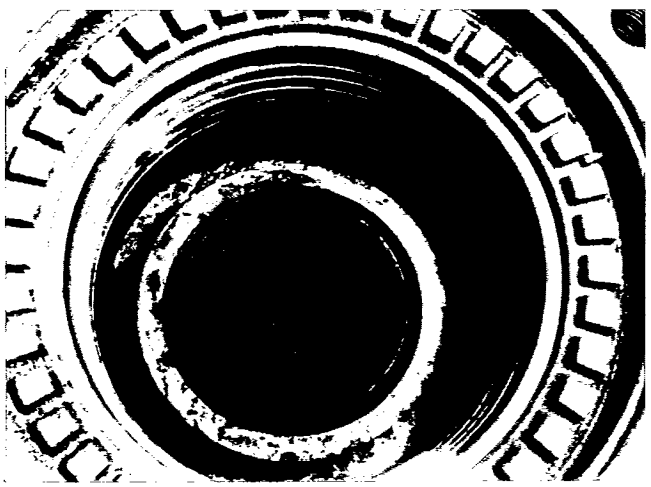


Figure 27—Inlet Nozzle Plugged With Coke and Catalyst After Nine Months in Reactor Vapor Line

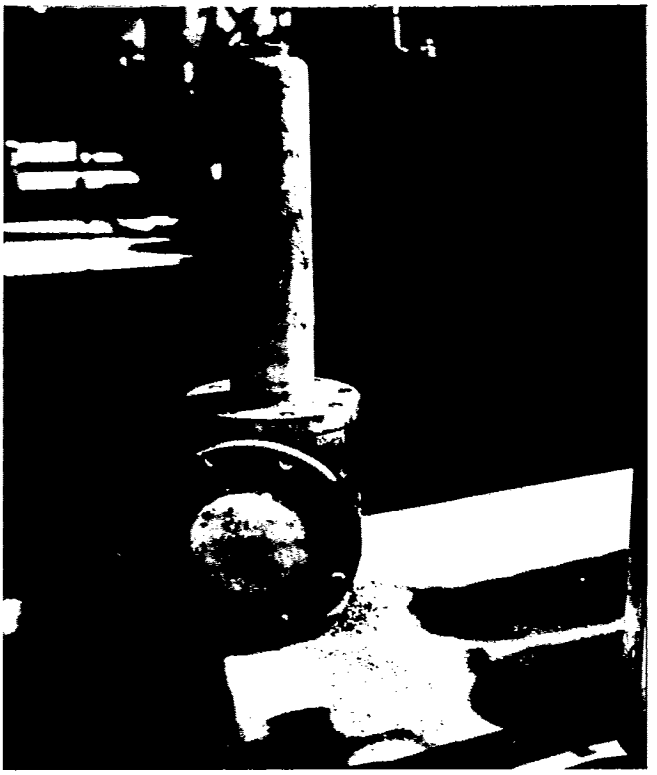


Figure 28—Outlet of Valve Plugged With Deposits From Other Valves in Common Discharge Header

When galling of the metal in the guiding surfaces is not due to corrosion or foreign particles, it is often due to valve chatter or flutter caused by improper piping at the valve inlet or outlet or by oversizing of the valve. Correction of improper piping at the valve inlet or outlet will usually stop galling. Galling of the guiding surfaces can also be caused by improper finishing of those surfaces. To reduce the chances of galling, they should be polished until they are as smooth as possible. Since the use of corrosion resistant metals in pressure relief valves is unavoidable and frequently leads to galling, all possible precautions against galling should be taken. Galling may be minimized by varying the materials and hardnesses of the contacting parts until the best arrangement is found.

Sticking of pressure relief valves may also be caused by poor alignment of the valve disk, which is usually due to improper cleaning of the contact surface between the guide and the body of the valve or misalignment of a gasket at assembly (see Part II of API Recommended Practice 520).

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Figure 29—Moving Parts of Valve Fouled With Iron Sulfide (FeS_2)

4.6 Misapplication of Materials

In general, the materials required for the pressure-relieving devices in a given service are determined by the temperature, pressure, and corrosion resistance requirements and the atmospheric conditions of the service. The selection of standard valves that meet those requirements and are appropriate for those conditions is advisable. Occasionally, however, severe corrosion or unusual pressure or temperature conditions in the process require special consideration. Manufacturers can usually supply valve designs and materials that suit special service. Catalogs show a wide choice of special materials for various chemical and temperature conditions.

The hydrogen sulfide (H_2S) attack on the carbon steel disk in Figure 18 and the chloride attack on the 18Cr-8Ni steel disk in Figure 19 exemplify the results of the misapplication of materials. Where service experience indicates that a selected valve type or material is not suitable for a given service condition, an immediate correction that will ensure dependable operation should be made. Great care should be taken to record the identity of special materials and the locations requiring them. An adequate system of records should provide the information needed for the repair or reconditioning of valves in special service and for developing optimum purchase specifications.

4.7 Improper Location, History, or Identification

If not installed at the exact location for which it was intended, a pressure relief valve loses its effectiveness and cannot provide the proper protection. To assist in the identification of valves and to provide information necessary for correct repairs and installation, a comprehensive set of specification and historical records should be maintained and referred to when valves are removed for inspection and repair. Most pressure relief valves have an identifying serial or shop number placed on the valve by the manufacturer or an identifying number tagged, stamped, or otherwise placed on the valve by the user. This identification specifies the location of the valve and, by reference to the specification record, its limitations and construction (see Section 8).

4.8 Rough Handling

4.8.1 GENERAL

Because of the difficulty in obtaining absolute tightness in most pressure-relieving devices, valves are manufactured according to a commercial tightness standard (see API Standard 527). Valves are checked for tightness in the manu-

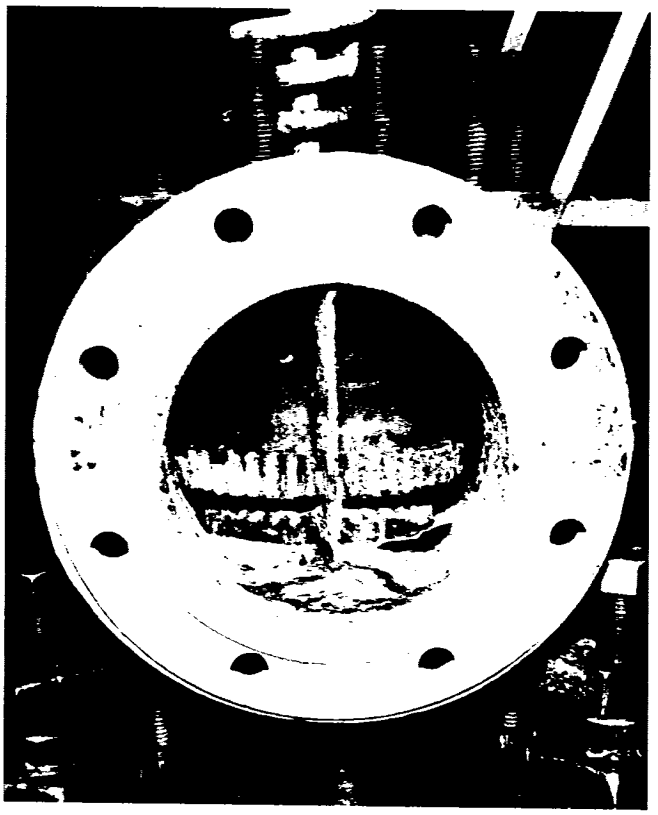


Figure 30—Valve Stuck Because of Iron Sulfide (FeS_2) Deposits

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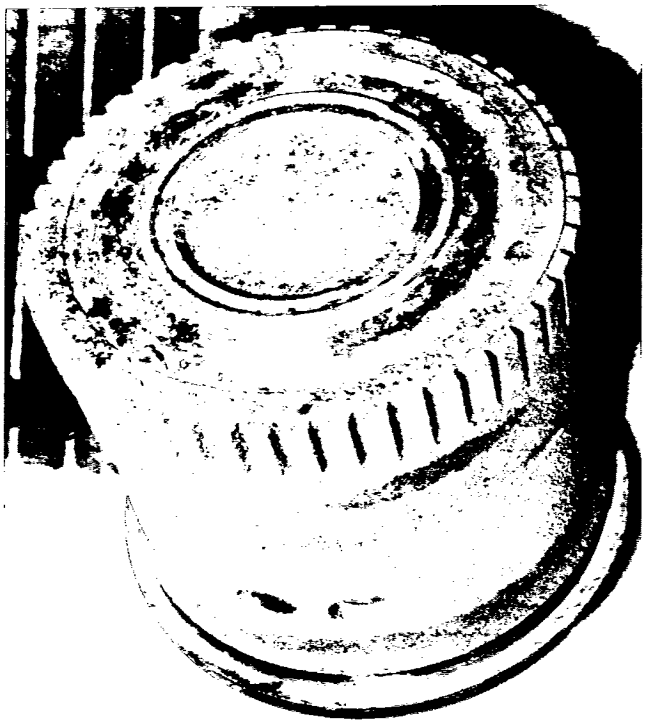


Figure 31—Disk Frozen in Guide Because of Buildup of Products of Corrosion in Sour-Oil Vapor Service

facturer's plant before they are shipped to the user. They are also usually checked by the user in the maintenance shop before initial use and after cleaning, repairing, or testing. Subsequent rough handling of the valve, however, can change the set pressure, damage lifting levers, destroy the tightness, and produce excessive leakage in the valve when in service. Rough handling can occur during shipment, maintenance, or installation of the valve.

4.8.2 DURING SHIPMENT

Because of their construction, most pressure relief valves have a sturdy appearance that may obscure the fact that they are delicate instruments with very low tolerances and sensitive dimensions. Accordingly, commercial carriers sometimes subject them to rough handling, which may cause the valves to leak excessively in service or during testing. This rough handling may also expose the valve inlet to dirt or other foreign particles that could damage the valve seating surface the first time the valve opens and cause leakage thereafter.

Relief valves should be shipped in an upright position —this is especially true of large relief valves and relief valves with low set pressures. When large, low-pressure relief valves are allowed to lay on their sides, the springs may not exert the same force all around the seating surfaces. The springs may vibrate and be damaged.

4.8.3 DURING MAINTENANCE

Pressure relief valve parts are usually precision items manufactured to extremely close tolerances. These tolerances can be destroyed by rough handling in the maintenance shop. Rough handling can also destroy the basic valve alignment on which the fine, exacting performance characteristics of the device primarily depend. Careful handling of the valve during all phases of maintenance is important. Both before and after repairs, rough handling of the completely assembled valve should be avoided. Before the valves leave the shop, valve inlets and outlets should be covered.

Rough handling during maintenance includes excessive back pressure, which should not be applied to a bellows valve during a maintenance-related test. The manufacturer's pressure rating of the bellows should be checked and followed.

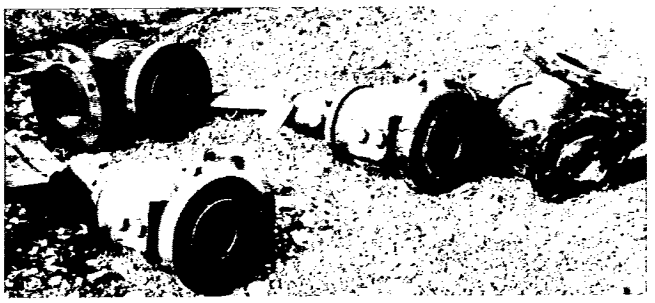
4.8.4 DURING INSTALLATION

Valve inlets and outlets should have been covered before the valves left the shop. If they were not covered when received for installation, provisions should be made to ensure that in the future they are covered before leaving the shop.

Rough handling of a pressure relief valve by personnel during installation may cause poor valve performance in service. Bumping or dropping the valve should be carefully avoided. The valves shown in Figure 32 were dropped from the bed of a truck after being repaired. As a result, they leaked once they were installed.

4.9 Improper Differential Between Operating and Set Pressures

The differential between operating and set pressures provides seat loading to keep the pressure relief valve tightly closed. Due to a variety of service conditions and valve de-



Note: The valve inlets and outlets should have been protected and covered.

Figure 32—Rough Handling of Valves Should Be Avoided

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signs, only general guidelines can be given for designing a system. NBBPVI NB-23 and Sections VI and VII of the ASME Code are useful references. However, individual applications and experience must ultimately be relied on. Although greater differentials between operating and set pressures promote trouble-free operation, they may also increase the cost of the equipment. Inspections should record operating experience and provide feedback to be considered in future design and remedial action.

4.10 Improper Piping Test Procedures

When hydrostatic tests of discharge piping are performed, blinds must be installed. Otherwise, results such as the following might occur:

a. The disk, spring, and body area on the discharge side of the valve are fouled.
b. The bellows of a balanced relief valve are damaged by excessive back pressure.

SECTION 5—FREQUENCY AND TIME OF INSPECTION

5.1 Inspection Frequency

5.1.1 GENERAL

To ensure proper performance, a definite time interval between inspections should be established for every pressure-relieving device on operating equipment. Depending on operating experiences, this interval may vary from one installation to another. The time interval should be sufficiently firm to ensure that the inspection is made, but it should also be flexible enough to permit revision as justified by past test records.

CAUTION: In evaluating inspection intervals, safety considerations are most important and supersede economic considerations.

In API 510, the subsection on pressure-relieving devices establishes a maximum interval between device inspections of 10 years. It also indicates that the intervals between pressure-relieving-device testings or inspections should be determined by the performance of the devices in the particular service concerned.

5.1.2 NORMAL BASIS

The interval between inspections of pressure-relieving devices should not exceed that necessary to maintain the device in satisfactory operating condition. The frequency of inspection is normally determined by operating experience in the various services involved. Obviously, the interval between inspections of a device in a corrosive and fouling service would be shorter than the interval required for the same device in a clean and nonfouling service. Likewise, more frequent inspection and testing should be provided for spring-loaded pressure relief valves subject to vibration, pulsating loads, low differential between set and operating pressures, and other circumstances leading to valve leakage and poor performance.

Where an adequate inspection history extending over a long period of time reflects consistent "as received" test results that coincide with the set pressure (see 7.2), where no change in service is to be made, and where no conflict in jurisdictional requirements exists, an increase in the test interval may be considered. Conversely, if the "as received" test results are erratic or vary significantly from the set pressure, the inspection interval should be decreased or suitable modifications to improve the performance (such as the installation of a rupture disk) should be made. If a valve fails to activate on the test block at the set pressure, it can be assumed that it would have failed to activate on the unit during an emergency.

Where corrosion, fouling, and other service conditions are not known and cannot be predicted with any degree of accuracy (as in new processes), the initial inspection should be accomplished as soon as practical after operations begin to establish a safe and suitable testing interval.

5.1.3 MANUFACTURER'S BASIS

Manufacturers of pressure-relieving devices are sometimes able to assist the user in establishing inspection and test intervals, especially if their designs contain features and components that require special consideration. For example, it may be necessary to inspect or replace certain parts, such as nonmetallic diaphragms in pilot-operated valves, at frequencies other than those required for the parts of conventional pressure relief valves. Rupture disks and bellows valves may also require special consideration. Manufacturers are familiar with the nature of the loading, stress levels, and operating limitations of their design and are able to suggest inspection intervals appropriate for their equipment.

5.1.4 OTHER BASES

In some instances, the required frequency of inspection and testing of pressure-relieving devices is established by regulatory bodies. Regulations applicable in the installation's locality should be investigated to avoid conflict.

5.2 Time of Inspection

5.2.1 GENERAL

If the operation of equipment has only been interrupted for normal shutdown, pressure-relieving devices should be

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inspected and tested according to schedules previously established.

Pressure and/or vacuum vent valves on atmospheric storage tanks should be inspected after installation but before the tank is hydrostatically tested or put into service. The items to check are covered in 7.3.2.

5.2.2 INSPECTION ON NEW INSTALLATIONS

All pressure relief valves and other automatic pressure-relieving devices that depend on a spring adjustment for proper functioning should be inspected and tested before they are installed on process equipment. This inspection is used to determine any damage or changes in factory adjustment due to shipping, confirm the set pressure, and initiate appropriate records. Records initiation should include the preparation of a complete specification record (see 8.4), the assignment of a company number, and the installation of company identification (metal tag, stencil, or other means; Figure 33 shows a tag with the company number, valve location, set pressure, and date of test).

Pressure and/or vacuum vent valves on atmospheric storage tanks should be inspected after installation but before the tank is hydrostatically tested or put into service. The items to be checked are covered in 7.3.2.

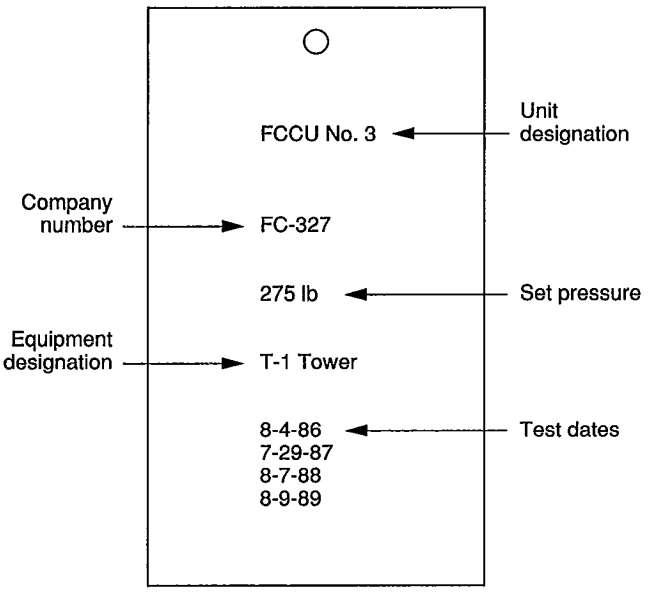
5.2.3 UNSCHEDULED INSPECTION

An inspection of a pressure-relieving device is considered on schedule if it follows the last inspection and repair by 90–110 percent of the established inspection interval, even if the last inspection and repair were unscheduled. Any other inspections should be approved by the inspector or by a person responsible for equipment safety. In the consideration of deviations from the schedule, the following should be considered:

a. *Service experience.* Historical records should be reviewed for the determination of whether safety would be compromised by the deviation.
b. *The ability to remove any block valves during operation.*
c. *The relation of the entire schedule to the planned unit shutdown.*
d. *Coordination with operations and with other work.*

CAUTION: Only where safety would not be impaired and regulatory requirements would be satisfied should economic considerations be the controlling factor.

Inspection schedules should allow deviations to correct any observed malfunctioning of pressure relief valves in service. If a valve fails to open at the set pressure, it certainly requires immediate attention. If it opens at the set pressure but fails to reseat properly, the urgency of inspection and repairs depends on the type of leakage, its environmental and human impact, the amount of leakage, and the characteristics of the leaking substance such as whether it's toxic, flammable, or fouling.



Note: Tag is 3 by 1½ by 1/16 inch and should be aluminum to resist corrosion and ease stenciling.

Figure 33—Identification Tag for Pressure-Relieving Device

5.2.4 INSPECTION DURING PLANNED SHUTDOWNS

The ideal time to inspect pressure relief valves is when the inspection least interferes with the process and maintenance manpower is readily available. These conditions prevail during planned shutdowns. All relief devices not equipped with block valves should be inspected at this time if an inspection would otherwise become due before the next scheduled shutdown. Even the relief devices with block valves should be inspected at this time to minimize process interruptions and avoid the increased risk of inspecting equipment in operation.

5.2.5 INSPECTION AFTER EXTENDED SHUTDOWNS

5.2.5.1 General

A pressure relief valve left on a unit during an extended shutdown should be inspected and tested before the resumption of operations. This inspection is necessary to ensure that corrosion, fouling, tampering, or other conditions or acts that would impede the proper performance of the device have not occurred during the shutdown. When a change in operating conditions is to follow the shutdown, the inspection interval should be reviewed.

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5.2.5.2 Inspection After Equipment Test and Inspection Downtime

After a test and inspection downtime of a unit or other process or operating equipment, pressure relief valves should be checked to ensure the following:

a. *The correct relief device was installed.*
b. *The company identification (such as a tag or stencil) provides means to establish the proper pressure setting for the equipment protected.*
c. *No gags, blinds, closed valves, or piping obstructions would prevent the valves from functioning properly.*
d. *Seals installed to protect the spring setting have not been broken.*
e. *Any bellows vents are open and clear, and the connected piping is routed to a safe location.*
f. *Upstream and downstream block valves are sealed or chained and locked in the proper position.* Devices that ensure that a block valve is in its proper position include locking plastic bands, car seals, chains and padlocks, and special locking devices made especially for certain types of block valves.

Note: A *car seal* is a metal-band, self-locking seal used to seal shut railroad boxcar doors. Once these seals are placed in position and closed, they lock and must be cut with wire cutters or physically broken to be removed.

g. *Vent stacks, discharge piping, and small nipples are properly supported to avoid breakage or leakage.* Inadequately supported or anchored nipples can be damaged during maintenance or by vibration.
h. *Valve body drains and vent stack drains are open.*
i. *Any lifting lever is operable and positioned properly.*
j. *A gauge installed as part of a rupture disk and pressure relief valve combination or a device for checking pressure between a pressure relief valve and a block valve is serviceable.*

5.2.6 VISUAL ON-STREAM INSPECTION

A visual on-stream inspection is more like a survey than a normal inspection and may be considered a control measure. It should ensure the following:

a. *The correct relief device was installed.*
b. *The company identification (such as a tag or stencil) provides means to establish the proper pressure setting for the equipment protected by the identified device.*
c. *If not clear from the records, that the established test interval has not been exceeded.*
d. *No gags, blinds, closed valves, or piping obstructions would prevent the valves from functioning properly.*
e. *Seals installed to protect the spring setting have not been broken.*
f. *The relief device does not leak.* Pressure relief valves that have opened in service frequently leak. Detection and correction of this leakage eliminates product loss and possible pollution and prevents fouling and subsequent sticking of the valve. If the valve is a bellows valve, the bellows vent should be checked for leakage.
g. *Any bellows vents are open and clear, and the connected piping is routed to a safe location.*
h. *Upstream and downstream block valves are sealed or chained and locked in the proper position.* Devices that ensure that a block valve is in its proper position include locking plastic bands, car seals, chains and padlocks, and special locking devices made especially for certain types of block valves.

Note: A *car seal* is a metal-band, self-locking seal used to seal shut railroad boxcar doors. Once these seals are placed in position and closed, they lock and must be cut with wire cutters or physically broken to be removed.

i. *Vent stacks, discharge piping, and small nipples are properly supported to avoid breakage or leakage.* Inadequately supported or anchored nipples can be damaged during maintenance and by vibration.
j. *Valve body drains and vent stack drains are open.*
k. *Any lifting lever is operable and positioned properly.*
l. *A gauge installed as part of a combination of a rupture disk and a pressure relief valve or a device for checking pressure between a pressure relief valve and a block valve is serviceable.*
m. *Any rupture disk is properly oriented.*

Although the interval selected for on-stream inspection should vary with circumstances and experience, a visual inspection that includes a check for leakage and vibration damage should follow each operation of a pressure relief valve. If sufficiently experienced to recognize any leakage or vibration damage, operating personnel assigned to the process unit could make these inspections.

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SECTION 6—PREPARATION, TOOLS, AND EQUIPMENT

6.1 Preparation

6.1.1 EQUIPMENT CHECKED FOR SAFETY

CAUTION: Before inspection and any repairs on pressure-relieving devices are executed, general precautions should be taken to maintain the safety of the equipment protected by the devices, especially if the equipment is in operation. When inspection and repairs on an operating unit are required, the unit operations should be normal and the proper authority and permits for the work should be obtained.

Before disconnecting relief valves, the connected piping and block valves should be checked to ensure that they are sufficiently supported. After reinstalling relief valves, the related piping should be checked to ensure that it is not imposing loads that would cause problems with the relief valve body such as distortion leading to in-service leakage.

CAUTION: Before relief lines are opened to the atmosphere, precautions should be taken to prevent the release of hydrogen sulfide (H_2S), hydrocarbons, and any other hazardous materials in the system, which might be able to flow from remote portions of the system out through relief line openings, and to prevent the ignition of iron sulfide (FeS_2) deposits in the piping when exposed to air.

In some cases a spool piece can be installed in place of the pressure relief valve to be inspected so that any excessive pressure that occurs while the valve is out can manually be safely released. When the safety valve is out of service, a bypass around the safety valve is sometimes used for depressurization.

If a block valve is to be provided between a vessel or other protected equipment and its only pressure relief device, it shall be so arranged that it can be locked or sealed open and shall only be closed by an authorized person who shall constantly monitor the vessel's operation while the valve remains closed and who shall lock or seal the stop valve in the open position before leaving (see the ASME Code, Appendix M).

CAUTION: The above procedure should be executed with the greatest care and planning and with the safety of the authorized person as the *top* priority.

When a block valve is provided between a vessel's only relief valve and the header the valve discharges into, the block valve shall be so arranged that it can be locked or sealed in either the open or closed position. The block valve shall be locked or sealed in either position by an authorized person only. If the block valve is to be closed while the vessel is in operation and the vessel has no other relieving device, the authorized person shall constantly monitor the operations and shall lock or seal the block valve in the open position before leaving. This valve should not be closed while the vessel is in operation unless a stop valve is on the inlet side of the safety relieving device and is first closed.

CAUTION: The above procedure should be executed with the greatest care and planning and with the safety of the authorized person as the *top* priority.

CAUTION: Before a pressure relief valve is inspected and possibly repaired in place or it is removed for inspection and any repairs while equipment is in operation, the following precautions should be taken:

a. By an authorized person *only*, the relief device should be isolated by closing any adjacent block valves upstream or downstream.
b. The space between the relief device and any adjacent block valve should be vented to a safe location to release trapped lading fluid and to determine whether the block valve is holding.
c. If no block valve is installed on the downstream side of a relief device discharging into a common header, a blind or other suitable means should be provided there to prevent discharge through the open outlet pipe in case one of the other relief devices opens.
d. Before a pressure relief valve is inspected and possibly repaired in place, a blind should be inserted between the pressure relief valve and any adjacent upstream block valve. If the pressure relief valve discharges into a common discharge header, a blind should be installed between the relief valve and the header.
e. After the relief device has been inspected, repaired if appropriate, and replaced if it had been removed, all blinds should be removed. The block valves on the inlet and outlet should be opened and locked or sealed in that position. Figure 34 shows a safety relief valve installation with the block valves sealed open. When used with a relief device, these block valves must have full pipe area through them to prevent flow restriction when the relief device opens.

CAUTION: If the equipment protected by the device to be inspected is out of service and open for other work before the inspection and possible repairs, the discharge side should be properly blinded if the valve is connected to a common discharge header that remains in service. In this situation, the blinding should be the only special safety precaution necessary.

6.1.2 VALVE IDENTIFICATION CORRECTLY INSTALLED

To minimize errors in the testing and handling of pressure relief valves in service (see Section 8), each should carry an identifying tag, stencil, plate, or other means to show its

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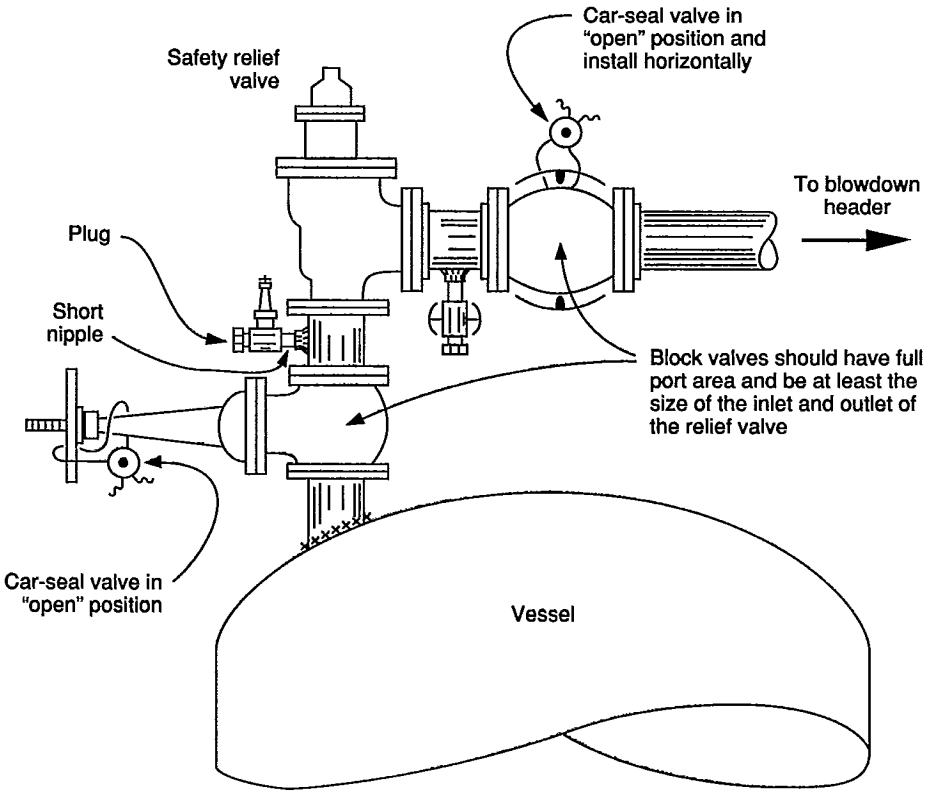


Figure 34—Block Valves on Safety-Relief-Valve Inlet and Outlet Sealed Open and Bleeders Installed to Depressure Inlet and Outlet

company equipment number. This number allows ready identification of the device's unit, the equipment that the device should be installed on, the device's set pressure, and the date of its last test (see Figure 33 for an example of an identifying tag). If a relief device cannot already be easily and correctly identified by a marking on it, it should be marked and identified as described above before it is removed from its equipment.

6.1.3 OPERATING CONDITIONS NOTED

A complete operating history of each pressure relief valve since its last inspection should be obtained and should include the following:

a. The average operating conditions.
b. The number and severity of upsets and their effect on the valve.
c. The extent of any leakage while in service.
d. Any other evidence of malfunctioning.

In addition, records of valve performance during previous runs should be checked to determine whether changes are needed in the valve materials or components or in the inspection interval.

6.1.4 CAREFUL AND MINIMAL REMOVAL OF DEVICE FROM SYSTEM IN OPERATION

CAUTION: The removal of a pressure relief valve from equipment in operation should be planned to minimize its duration. Relief devices are safety devices and should be treated as such. The precautionary steps in 6.1.1 should be followed.

Before the valve is removed, the proper tools, such as wrenches and hoists, should be gathered. The valves should not be handled roughly, and the bolting of body flanges should not be hammered. Rough handling of the valve or hammering on the body flange bolting could affect the valve setting so that testing and inspection results would not give a true picture of valve performance.

In rupture disk removal, the removal of parts other than the outlet pipe and the flanges holding the disk may be unnecessary.

6.2 Shop Test Block

6.2.1 NEED AND FUNCTION OF TEST BLOCK

After a pressure relief valve is removed from service, it is usually taken to the shop for inspection and repair. An im-

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portant phase of maintenance is testing to determine the set pressure and tightness of the valve "as received" and after its overhaul. The testing is usually performed on a test block with facilities for applying pressure to a valve and indicating the pressure applied. Most test blocks have facilities for testing with either air or water to simulate, as closely as possible, the media handled by safety and relief valves, respectively. Bottled nitrogen may be used instead, especially for high-pressure valves. See Sections I, IV, and VII of the ASME Code for requirements for setting safety valves and safety relief valves in steam service.

The test block and its supporting facilities are necessary for the maintenance of pressure relief valves. It is practically impossible to make accurate adjustments on these devices without some method of measuring their performance. The shop test block, unfortunately, does not duplicate field conditions exactly. Thus, the amount of liquid or gas that it can discharge is limited, and it is not generally practical to measure relieving capacity or blowdown. Also, test stands with insufficient surge volume may fail to cause a distinct pop, and an inaccurate set pressure may result. However, if properly functioning, the shop test block gives good indications of the pressure at which the valve will open and its tightness.

CAUTION: For safety, the valve discharge nozzle must be positioned to prevent exposure of personnel to a sudden blast of air, water, or other projectiles from the valve. Ear protection may also be required for personnel working in the test area.

If a pressure relief valve is dirty and popping the valve would damage its seats, the "as received" pop test may be waived. If the test is waived, reduction of the valve's test interval should be considered.

6.2.2 TESTING WITH AIR

Most test blocks are designed to test pressure relief valves with air because it is a safe and readily available medium. Air is compressible and causes reaction-type valves to relieve with a short pop and closely approximates operating conditions for pressure relief valves in hydrocarbon or other gas services. The air test is generally used to test safety, relief, and safety relief valves for set pressure and valve tightness.

The arrangement to detect leakage during the air test depends on the construction of the valve. Blinding of the valve discharge is usually required. Leakage may be detected qualitatively by placing a thin membrane, such as a wet paper towel, over the outlet and noting any bulging. This is not a rigid test and is not intended to be used as a commercial standard tightness test. A quantitative measurement may be made by trapping the leakage and conducting it through a tube submerged in water. Figure 35 shows the standard equipment used to determine leakage rates in API Standard 527. Leaks can also be detected with ultrasonic sound detection equipment.

6.2.3 TESTING WITH WATER

Test blocks may include facilities that test relief valves with a liquid test medium such as water. Water is safe and inexpensive and may allow a close simulation of operating conditions. Because very small water leaks are not readily detected, a water test is usually limited to measuring the set pressure. Leakage or tightness tests are usually made with air (see 6.2.2). However, leakage tests may instead be made with water as described in API Standard 527.

6.2.4 DESCRIPTION OF TEST BLOCK

The test block is the equipment assembled to test pressure relief valves for set pressure and tightness. It is used often and must be readily available on short notice. Test block designs vary widely and are even offered as packaged equipment by some manufacturers. The schematic arrangement in Figure 36 shows the essential elements of and instructions for a test block that uses air as the testing medium. Where air pressure is unavailable, water systems may instead be used to test relief valves.

The air-system test block includes a compressor or other source of high-pressure air, a supply reservoir, a test drum or surge tank large enough to accumulate enough air to cause the valves to open sharply at the set pressure, and the piping, pressure gauges, valves, and other instrumentation necessary to control the tests. The water-system test block usually includes a positive displacement pump that with a reasonable flow of water develops high pressures and the piping, valves, and other instrumentation necessary to control the tests.

Both the air-system test block and the water-system test block use a manifold. The wide range of flange sizes on a manifold allows it to test many different valves. To cover the wide range of pressures usually required to test pressure relief valves, several precision-calibrated pressure gauges are connected with the manifold. These pressure gauges should be routinely calibrated, and a progressive calibration record should be maintained.

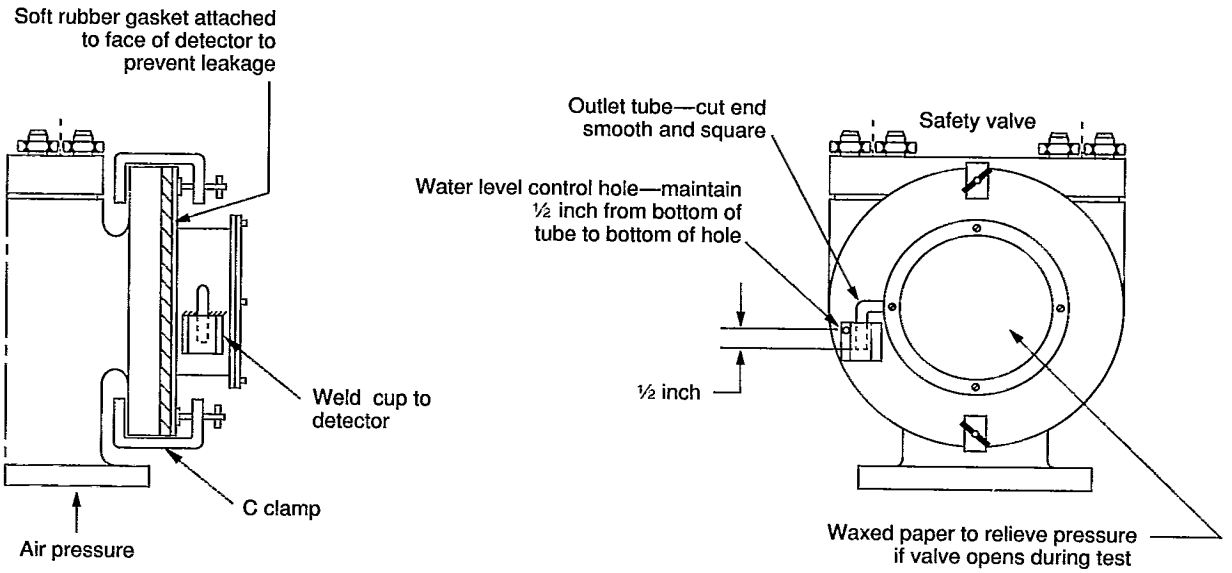
6.3 Tools and Spare Parts

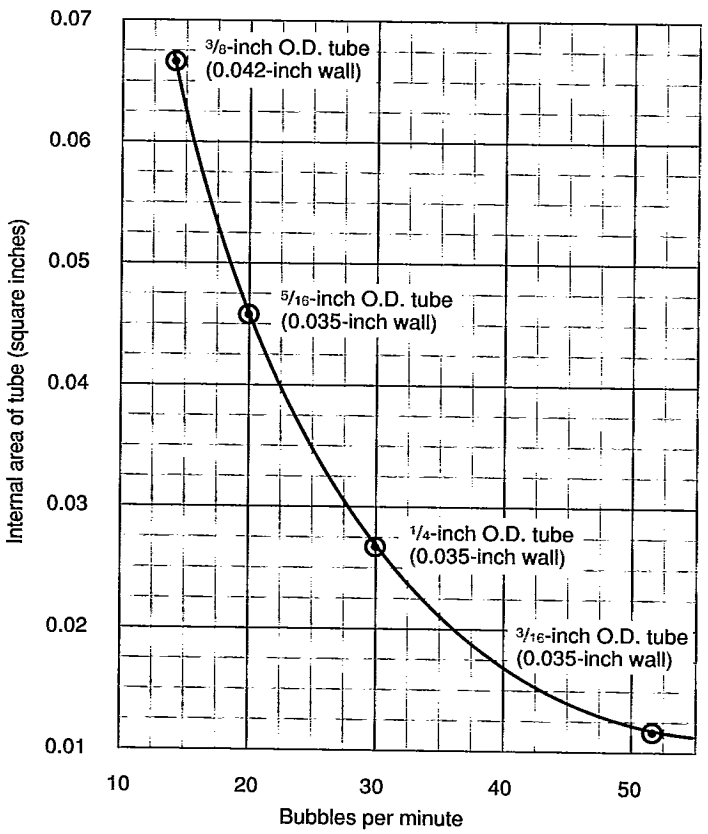
6.3.1 TOOLS

In addition to the leak detector (bubble test facilities) discussed in 6.2.2 and illustrated in Figure 35 and the test block discussed in 6.2.4 and illustrated in Figure 36, the following tools needed to test and repair pressure relief valves should be kept in the machine shop:

a. Wrenches.
b. Any special tools recommended by the manufacturers of the valves.

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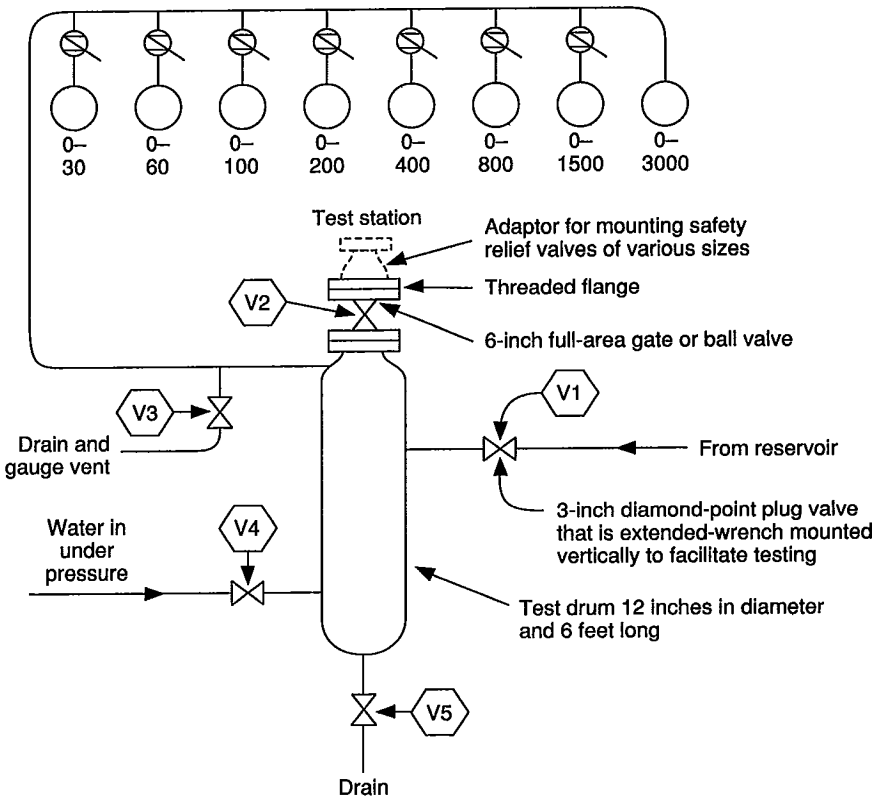


Leakage rates corresponding to 0.3 cubic feet in 24 hours

Notes: When using this detector, all other valve outlets must be closed. Leakage should be sought at 90 percent of the set pressure. Bubbles should be counted after they appear at uniform rate.

Figure 35—Safety-Valve and Relief-Valve Leak Detector

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Notes on Construction:

1. This layout uses the available air supply at the highest pressure possible. If required, the pressure can be raised further by inserting water that is under pressure into the test drum until the desired pressure is reached.
2. A single test drum is shown. Duplicate stations for flanged valves can be added if desired. Another duplicate station with a smaller test drum is sometimes desirable for testing small valves with screwed connections.
3. Flanged valves are to be secured to test stations by bolting, clamping, or use of a pneumatic clamping device.
4. Line from reservoir to test drum is to be designed for minimum pressure drop to allow reservoir volume to hold up test drum pressure when needed.
5. Test drum pressure and piping are to be made of oxidation-resistant materials.

Notes on Operation:

1. When test station is not in use, valves V1, V2, V4, and V5 should be closed. Valve V3 should be opened to prevent possible buildup pressure in the test drum if valve V1 should leak.
2. Before testing the first valve, the test drum should be blown to remove any accumulation of dust or sediment that might blow through the safety relief valve and damage the seats. To blow the drum, close valve V3, open valve V2, and release air through the drum by opening and closing valve V1.
3. Close valve V2.
4. Secure safety relief valve to test station.
5. Open valve V2.
6. If safety-valve set pressure is lower than available air pressure, slowly increase pressure through valve V1 until safety relief valve pops. Then close valve V1. If safety-valve set pressure is higher than available air pressure, open valve V1 and fill test drum with maximum air pressure available. Then close valve V1. Open valve V4 and increase pressure by inserting water that is under pressure until safety relief valve pops. Then close valve V4 and drain water from drum by opening and closing valve V5.
7. If necessary, adjust valve spring so that safety relief valve pops at the required set pressure.
8. Vent the test drum to 90 percent of the set pressure.
9. Test safety relief valve for leakage.
10. After satisfactory test, close valve V2.
11. Remove safety relief valve from test rack. Loosen bolts or clamps slowly to allow pressure in adapter and valve nozzle to escape.
12. Vent test drum through V3 to approximately 75 percent of the set pressure of the next valve to be tested. Repeat 4 thorough 11.
13. If another valve is not to be tested immediately, leave test station as specified in 1.

Figure 36—Typical Safety-Valve and Relief-Valve Test Block Using Air as Test Medium

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6.3.2 SPARE PARTS

An adequate supply of spare parts should be readily available to repair the valves in service. Spare parts should be correctly identified and properly protected against damage from rough handling or corrosion. A list of spare parts should be developed from past experience with the valves and any recommendation made by the valve manufacturers and contractors. Where vendor stocking programs are used, the programs' spare parts lists should be checked against the list being developed. The vendor stocking agent and the user should agree on the parts to be stocked. A typical spare parts list might include the following:

a. Springs.
b. Gaskets.
c. Disks.
d. Nozzles.
e. Bolts.
f. Bellows.
g. Frangible disks.
h. Nonmetallic diaphragms.
i. Complete relief valves of most commonly used sizes.
j. Pilot kits.
k. O rings.

SECTION 7—INSPECTION, TRANSPORTATION, TESTING, MAINTENANCE, AND SETTING

7.1 Initial Inspection of Pressure Relief Valves and Adjacent Piping and Transportation of Valves to Shop

7.1.1 INSPECTION OF VALVES

Many types of deposits or corrosion products in a pressure relief valve may be loose and may drop out during transportation of the valve to the shop for inspection, testing, maintenance, and resetting. As soon as a valve has been removed from the system, a visual inspection of it should be made. Figure 37 shows sulfur deposits in a valve. When fouling is a frequent problem, it may be prudent to collect samples for testing and to record deposit locations and appearances. Any obstructions in the valve should be recorded and removed.

CAUTION: Valves that have been exposed to materials hazardous to humans or that may contain material that could be an auto-ignition source should be handled with special precautions. Some precautions follow:

a. Valves that might contain pyrophoric materials should be kept wet to prevent ignition.
b. Valves in hydrofluoric acid (HF) *must* be correctly neutralized immediately after removal. Even after neutralization, the safety precautions indicated by the material safety data sheets and other appropriate sources of handling information must be taken.

Rupture disks are sometimes used to protect other pressure-relieving devices from corrosion. Normally in this case, a rupture disk cannot be inspected without being removed. Therefore, inspection of the disk should be part of the routine developed for inspection of the pressure relief valve (see 7.5).

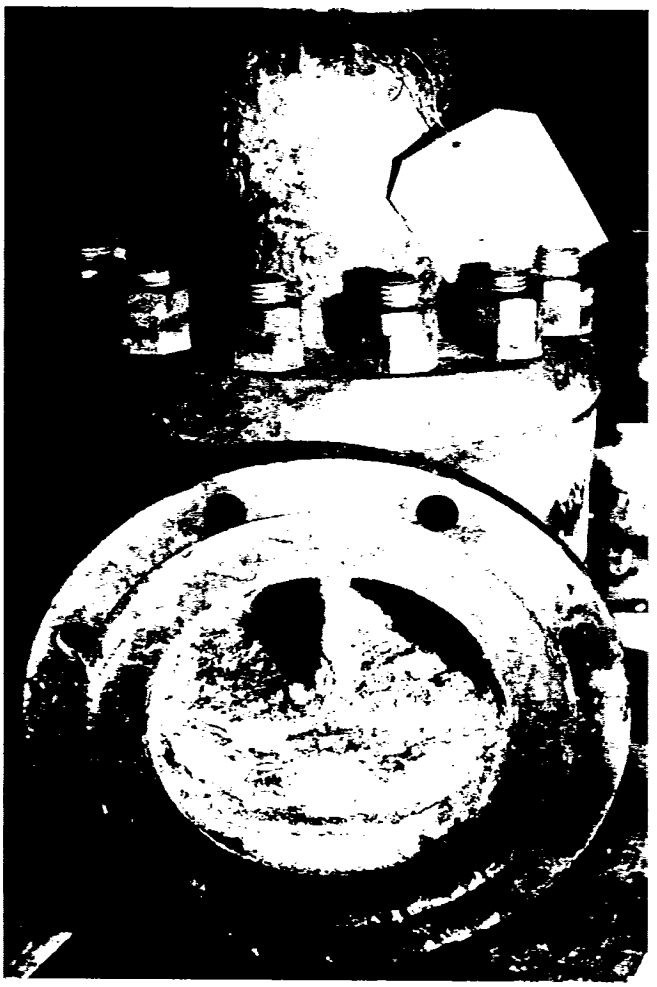


Figure 37—Sulfur Deposits in Body of Valve on Way to Repair Shop

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7.1.2 INSPECTION OF ADJACENT INLET AND OUTLET PIPING

When a pressure relief valve is removed from service, the upstream and downstream piping is often open and available for inspection. However, where block valves are closed to enable removal of relief devices from equipment during operation, it is usually impossible to directly inspect this piping. It should instead be directly inspected once the operating equipment is out of service. Before then, radiographs of the piping can indicate any major fouling or stoppage.

Inspection of the piping at the pressure relief valve will often indicate the condition of the process piping whose interior is not visible. Piping should be checked for corrosion, indications of thinning, and deposits that could interfere with valve operation. The character of the deposits may indicate the cause of any leakage from the valve in a closed system. Figures 27 through 31 and Figure 37 show fouling deposits in pressure relief valves.

7.1.3 TRANSPORTATION OF VALVES TO SHOP

After a pressure relief valve is removed from its installation, it should be transported to the shop for inspection, testing, maintenance, and resetting. Before moving the valve, flange faces on the valve (and flanges on the piping) should be protected so that gasket faces are not damaged. The valves should be transported in the upright position. Large valves should be handled with suitable rigging equipment to prevent them from being dropped or roughly handled. Upon arrival at the shop, the valves should not be handled or stored in a careless manner. Where practical, they should be segregated and stored away from heavy pipe fittings or other types of valves.

CAUTION: Because rough handling can change the pressure setting of a pressure relief valve or deform its parts so that they cannot be set properly, careful handling in transit is very important. Pressure relief valves should be treated as delicate instruments because their accurate functioning is critical to safe plant operation.

7.2 Inspection, Testing, Maintenance, and Setting of Pressure Relief Valves in the Shop

7.2.1 POP TESTING OF VALVE "AS RECEIVED"

When a valve is first received in the shop, it should be placed on the test stand and pressured to find its "as received" relieving pressure (pop pressure).

7.2.2 VISUAL INSPECTION

After the test pop, a valve should be visually inspected for an estimate of its condition when removed from service. This inspection should be made by the shop's relief valve repair mechanic unless unusual corrosion, deposits, or conditions are noted in the pressure relief valve (see first Caution below). The results of this inspection should be noted on appropriate forms. Points that should be checked include the following:

a. The flanges, for evidence of pitting, roughening, or decreases in the width of seating surfaces.
b. The springs, for evidence of corrosion or cracking and for the correct pressure range at the valve's operating pressure and temperature.
c. The bellows, if the valve is of the bellows type.
d. The positions of the set screws and openings in the bonnet.
e. The inlet and outlet nozzles, for evidence of deposits of foreign material or corrosion.
f. The external surfaces, for any indication of a corrosive atmosphere or of mechanical damage.
g. The body wall thickness.
h. Valve components and materials, for a match with the information on the identification tag and specification card.
i. The pilots and associated parts.

CAUTION: When unusual corrosion, deposits, or conditions are noted in the pressure relief valve, a refinery inspector should make the inspection.

CAUTION: If the pressure relief valve is from equipment handling hazardous materials, caution should be exercised during the inspection.

7.2.3 DETERMINING "AS RECEIVED" RELIEVING PRESSURE

Before the valve is dismantled, the pop pressure of the valve when removed from service should be estimated. Although procedures for determining relieving pressures vary with local plant practices, generally the operator determines relieving pressure by mounting the pressure relief valve on the test block, slowly increasing the pressure on the valve, watching the appropriate pressure gauge, and noting the point at which the valve relieves the pressure (pops). The "as received" test pressure should be recorded for review and facilitation of any necessary corrective action.

If initially the valve opens at the set pressure, no further testing to determine the "as received" relieving pressure is needed. If the initial pop is at a pressure higher than the set pressure, the valve should be tested a second time. If it then pops near the set pressure, the valve was probably stuck because of deposits. If instead on the second try the valve does not pop near the set pressure, either the valve setting was originally in error or it was changed during operation. If the initial pop is at a pressure lower than the set pressure, either the spring has become weakened or the setting was changed during operation.

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CAUTION: If the valve when received is extremely fouled and dirty and the "as received" popping of the valve may damage the valve's seats, the user may waive the "as received" test pop and instead reduce the inspection interval. If all goes well, after reducing the valve's interval, the valve will be clean at the next inspection. If it is not then clean, the test interval should again be shortened or other measures should be taken to reduce the fouling.

7.2.4 DISMANTLING OF VALVE

After the valve is received and its pop pressure is checked, it is ready for thorough shop inspection and repair. The valve should be carefully dismantled in accordance with the manufacturer's manuals and recommendations. To avoid a flash due to sparks created by the dismantling operations, valves in light hydrocarbon service should before dismantling be cleaned thoroughly with the solution that is recommended by chemical cleaning agencies and whose chemicals are compatible with the valve material. Proper facilities should be available for segregation of the valve parts as the valve is dismantled. At each stage in the dismantling process, the various parts of the valve should be visually inspected for evidence of wear and corrosion. The valve stem, guide, disk, and nozzle require visual inspection. The bellows in balanced valves should be checked for cracks or other failures that may affect performance (see Figure 21).

7.2.5 CLEANING AND INSPECTION OF PARTS

To keep the parts of each valve separate from those of other valves, the valve parts should be properly marked and segregated. Then they should be cleaned thoroughly. The valve parts that most often require cleaning are the nozzles, springs, and seats. Deposits that are difficult to remove should be cleaned with solvents, brushed with wire, or carefully scraped.

After being cleaned, the separate parts should be checked carefully. An important step, the checking of the valve components should be done carefully with the proper equipment for measuring valve dimensions and with frequent reference to the proper drawings and literature.

The components should be checked for wear and corrosion. Seating surfaces on the disk and nozzle should be inspected for roughness or damage (see Figures 23 and 24), which might result in valve leakage. They should also be checked with appropriate seat gauges to assure that neither wear nor previous machining has made the seat dimensions exceed the manufacturer's tolerances. Seat flatness can be checked with suitable lap rings recommended by the manufacturer, optical flats, or other suitable inspection devices. If the appropriate equipment is available, the springs should be checked for the proper force. The springs should also be checked for cracking or deformation. The fit between the guide and disk or disk holder should be checked for proper clearance and visually inspected for evidence of scoring. The nozzle should be checked for obstructions and deformation. Bellows should be checked for leaks, cracks, or thin spots that may develop into leaks. In addition, if the bellows has collapsed, it has probably been subjected to back pressure greater than its design pressure. High back pressure may be due to downstream flare-piping restrictions, which are created by deposits, or to higher relief flows than used in the original design. The cause should be determined, and corrective action should be taken.

7.2.6 RECONDITIONING AND REPLACEMENT OF PARTS

Parts that are worn or damaged should be replaced or reconditioned. Damaged springs and bellows should be replaced. Spare parts for a particular pressure relief valve may be obtained from its manufacturer. The valve body, flanges, and bonnet may be reconditioned by means suitable for repairs to other pressure-containing parts of similar material. If evidence of wear or damage is found on the disk or nozzle, their seating surfaces may be machined (see Figure 17) or lapped. The manufacturer's recommendations for lapping seats should be followed.

7.2.7 REASSEMBLY OF VALVE

After the valve has been inspected and its parts have been reconditioned or replaced, it should be reassembled in accordance with the manufacturer's instructions. The nozzle and disk seating surfaces should *not* be oiled. Clearances between assembled parts should be checked. In accordance with the manufacturer's instructions, the spring should be adjusted to pop as close to the desired set pressure as possible. Blowdown rings should be set carefully and accurately in accordance with the manufacturer's recommendations for both vapor and liquid services, and the settings should be noted for future reference. Because usually a test block cannot measure the actual blowdown, past performance should be evaluated to estimate any necessary adjustment.

7.2.8 SETTING OF VALVE TO SET PRESSURE

After the valve has been reconditioned and reassembled, its spring should be adjusted for the last time to ensure the valve will pop at the required set pressure. Although test procedures will vary with local plant practice, the valve is generally mounted on the test block and air pressure is increased slowly until the valve pops. The manufacturer's recommendations should guide the adjustment of the spring to the correct setting. If a new set pressure is required, the manufacturer's limits for adjustment of the spring must not be exceeded and applicable code requirements must be observed. If necessary, a different spring should be provided.

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After the valve has been adjusted, it should be popped at least once to prove the accuracy of the setting. Normally, the deviation of the pop pressure from the set pressure should not exceed ±2 pounds per square inch (±13.8 kilopascals) for pressures less than or equal to 70 pounds per square inch (483 kilopascals) or ±3 percent for pressures greater than 70 pounds per square inch (483 kilopascals) [see Section VIII, Division 1, paragraph UG 134(d)(1), of the ASME Code]. For pressure relief valves that comply with Section VIII, Division 1, paragraph UG 125(c)(3), of the ASME Code, the deviation shall not be less than 0 percent or greater than +10 percent. Any allowance for hot setting should be made in accordance with the manufacturer's data. Any adjustment to set pressure (cold differential test pressure) required to compensate for in-service back pressure, service temperature, or test media should be made in accordance with the manufacturer's valve specification data.

When the pressure is tested with water, it should be raised slowly to the required setting. The discharge should be observed for evidence of leakage, or the test gauge should be observed for a momentary drop in pressure. Attainment of the set pressure is usually indicated by a small continuous stream of water from the valve discharge. The pressure at which the valve releases should be within the tolerances noted above before the valve is approved for service.

7.2.9 CHECKING OF VALVE FOR TIGHTNESS

Once the valve is set to pop at its set pressure, it should be checked for leakage. On the test block, it can be tested for tightness by increasing the pressure on the valve to 90 percent of the set pressure and observing the discharge side of the valve for evidence of leakage. Methods of determining leakage are covered in 6.2.2 and Figure 35.

CAUTION: Leakage from pressure relief valves should be minimized because of the hazards to the environment, personnel, and equipment and because leakage leads to fouled and inoperable valves and to product loss.

7.2.10 COMPLETION OF NECESSARY RECORDS

All necessary records should be completed before a valve is placed back into service. By helping to determine when to replace the components of the valve and when to retire it, the records are critical to its effective future use. They form the historical record of the conditions and services under which the valve operated. Retention of maintenance and test records may be required by governmental regulations.

7.3 Inspection, Testing, Maintenance, and Setting of Pressure Relief Valves on Equipment

7.3.1 PROCESS UNITS

It is generally more economical and effective to inspect, test, maintain, and set a pressure relief valve at the shop at the required intervals than to do so on its equipment. However, when a valve operates in clean, low-temperature, low-pressure service, experience may indicate that inspection of the valve while on the equipment is safe and suitable. When checking a bleeder between the pressure relief valve and the block valve ahead of it confirms that the block valve holds tight, the bonnet of the pressure relief valve may be removed for immediate inspection, testing, and any minor repairs. When major repairs are indicated, the valve should be sent to the shop.

The pressure relief valve may be tested for set pressure and leakage with an inert-gas testing medium through the bleeder. Although this method is inferior to the test block procedure discussed in 6.2, it may be more appropriate in some services.

Reverse-buckling rupture disks may be used to facilitate and allow more frequent on-stream testing of pressure relief valves. For such testing, the section between the rupture disk and the pressure relief valve is generally pressured with an inert-gas testing medium. Since visual inspection and testing is not carried out in the shop when using this procedure, the rupture disk must be inspected and replaced on a regular basis.

A pressure relief valve may be tested on stream with specialized equipment by hydraulically lifting the valve stem while the valve remains in service. While a displacement transducer records the exact moment of lift and reseat, a load cell measures the force needed to move the stem. A record of both the valve lift and the reseat pressure may be printed. This method may not be accepted by all jurisdictions as a valid method of either verifying or adjusting valve set pressures.

CAUTION: Although it may be satisfactory for a clean-service pressure relief valve, this method of checking the set pressure of a pressure relief valve does not satisfy the need to remove any other valve for physical inspection and verification that all of its components are in satisfactory and safe working condition.

7.3.2 STORAGE TANKS

The relieving devices on tanks may be separated into pressure and/or vacuum relief devices on atmospheric tanks and pressure relief valves on pressure storage vessels. These devices are usually inspected, tested, maintained, and set while the equipment is in service.

The inspection, testing, maintenance, and setting of relieving devices on pressure storage tanks is similar to those of pressure relief valves on process equipment, as described in 7.2.

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Pressure and/or vacuum vent valves on atmospheric tanks are designed to vent air and vapor from the tank during filling operations and to admit air when the tank is drawn down (see Figure 9). Because pressure and/or vacuum vent valves are in continuous service, they are prone to failure by sticking unless examined periodically. Where temperatures fall below freezing, the devices should be checked during the cold period to ensure that the pallets do not stick because of icing. These pallets are usually weight loaded. The inspection of each vent valve in place should include the checking of the discharge opening for obstructions. The top of the valve should be removed and the pallets checked for freedom of movement. Seats should be checked to ensure that there is no sticking or leakage, since the forces actuating the valve are small. If the valve has a flame arrester on the inlet nozzle, it should be inspected for excessive fouling or plugging. If necessary, it should be removed for cleaning.

CAUTION: Inspection of the flame arrester in service is important because any malfunction may result in the loss of expensive equipment.

Inspection, testing, maintenance, and setting of pressure and/or vacuum vent valves should include the following special steps:

a. *Sticking should be corrected and prevented.* The disks (pallets) of the devices should be checked for sticking. If the disks are stuck, the product's effect on the seal material and on the disk material should be investigated. If necessary the seal material and the disk material should be changed.
b. *The disk should be checked and maintained.* Once a disk is removed, it should be cleaned, and its mass should be determined. Check the mass against the mass required for the correct relieving pressure of the device. [The setting of a pressure-vacuum device depends on the mass of the disk compared to the area of the opening covered by the disk. Set pressure is usually a standard ½ ounce per square inch (0.43 kilopascal), but may go as high as 24 ounces per square inch (10.34 kilopascals).] If the mass is not correct, mass must be added or removed until the correct mass has been achieved. Disk condition and serviceability should be checked, and unusable disks should be replaced.
c. *The seats and pallets of the disk should be checked and cleaned.*
d. *The gaskets at the disk seating areas must be checked and, if necessary, replaced.*
e. *The protective screens must be checked for serviceability and, if necessary, renewed.*
f. *Hinges and hinge pins must be checked for operability and, as necessary, serviced, lubricated, and replaced.*
g. *Any special coating used internally or externally on the body should be checked and, if unserviceable, replaced.*
h. *The hood should be inspected and, if unserviceable, replaced.*
i. *The bolts should be checked and, as required, replaced.*
j. *The procedures in 7.2 should be applied to the pressure and/or vacuum vent valves.*

7.3.3 BOILERS

Although safety valves on steam boilers are similar in construction and operation to relieving devices on process equipment, they are designed and installed in accordance with local, state, and federal regulations and power codes. Inspection of these devices should be in accordance with the regulatory requirements and the manufacturers' recommendations. Company practices may be used to establish an inspection policy if they do not conflict with or compromise the intent of any regulatory requirements.

Because Section I of the ASME Code does not permit block valves under boiler safety valves, each safety valve must be tested periodically by raising the steam pressure until the valve pops. Precision-calibrated pressure gauges should be used to determine the pressure at which the valve pops. The accumulation and blowdown should also be noted. The ASME Code also requires that boiler safety valves have a substantial lifting device by which the valve disk may be lifted from its seat when the working pressure on the boiler is at least 75 percent of the full working pressure, so that checking for the freedom of moving parts to operate is feasible.

Instead of being tested on the boiler, safety valves may be removed and tested at regular intervals, which may be determined by local jurisdictional requirements. The testing must meet the requirements of applicable codes, laws, and any other regulations of the jurisdictions involved. Usually testing for set pressure and blowdown with steam is required.

Some regulatory agencies allow on-stream testing of steam safety valves with special equipment that hydraulically lifts the valve stem while the valve remains in service. While a displacement transducer records the exact moment of lift and reseat, a load cell measures the force needed to move the stem. A record of both the valve lift and the reseat pressure may be printed.

CAUTION: This method of checking the set pressure and functioning of a safety valve satisfies the need to raise the boiler's pressure above its normal operating pressure, but does not satisfy the need to remove the valve for physical inspection and verification that all of its components are in satisfactory and safe working condition.

7.4 Inspection, Testing, Maintenance, and Setting of Pilot–Operated Safety Relief Valves

Although a variety of valve types operate with pilots, in every case pressure in the system activates a mechanism that is independent of the basic relief valve and that opens

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the valve to full capacity. Pilot valves are usually small spring-loaded valves that unload pressure above the basic valve diaphragm (see Figure 6) or valve piston (see Figures 7 and 8). Separate air-operated or electrically operated pilots are not considered reliable enough to be used as pressure-relieving devices.

Inspection, testing, maintenance, and setting of pilot-operated valves may be divided into two separate phases: the pilot mechanism phase and the basic valve phase. With test connections, the set pressure of some types of valves may be accurately tested while the valve is in service. If there is no block valve under the basic valve, it may be inspected and repaired only while the vessel is out of service.

Because of the many types of pilot-operated valves available, the valve manufacturer's recommendations for inspection and repair should be consulted and followed. Manufacturers of diaphragm valves (see Figure 6) frequently recommend semiannual inspection of the diaphragms and other components.

7.5 Inspection and Replacement of Rupture Disks

Rupture disks are used in services where a large relieving capacity is required to give a rapid drop in pressure. The disks are made from a wide variety of metallic and nonmetallic material. Rupture disks may be inspected visually. Inspection should include a check of flanges for leakage and a check of the disk for fatigue, corrosion, failure, and buildup of coke or other foreign material that could adversely affect the disk's performance.

If a disk's manufacturer specifies a limited bolting torque procedure and it is removed for visual inspection, it should be replaced. Once the bolting has been loosened, it is unlikely the disk can be reseated in exactly the same position and condition as before.

Because they cannot be tested, rupture disks should be replaced on a regular schedule based on their application, the manufacturer's recommendations, and past experience. If a block valve is located ahead of the disk, the block valve should be locked open during operation. If replacement of the disk is necessary, the block valve should be locked closed until disk installation has been accomplished.

CAUTION: The block valve may only be closed in accordance with 6.1.1.

Rupture disks are also used to protect other pressure-relieving devices from corrosion. Normally in this case, a rupture disk cannot be inspected without being removed. Therefore, inspection of the disk should be part of the routine developed for inspection of the pressure relief valve.

A rupture disk installed upstream of the relief valve to prevent valve corrosion is normally set to burst either at the set pressure of the pressure relief valve it is protecting or at a pressure 3–5 percent less than the set pressure. Between such a rupture disk and the relief valve should be installed tell nozzles with pressure gauges and excess flow valves or other devices to detect a possible disk leakage or rupture as required by applicable codes. See Section VIII of the ASME Code for venting requirements.

To prevent corrosion of or minimize leakage from a relief valve, a rupture disk may be installed downstream of the valve. Rupture disks are usually set at a pressure less than that of the relief valve. The space between the valve disk and the rupture disk shall be vented or drained to prevent the accumulation of pressure due to a small leak from the valve. See Section VIII of the ASME Code for other requirements.

CAUTION: The relief valve must be designed so that it will not fail to open at its proper pressure setting regardless of any back pressure accumulated between the valve disk and the rupture disk. If back pressure builds up between an ordinary spring-loaded safety relief valve and a rupture disk, the valve will not open at its set pressure. In this case, the valve should be replaced by a specially designed valve such as a diaphragm valve or a valve equipped with a bellows above the disk.

SECTION 8—RECORDS AND REPORTS

8.1 Objective

A suitable system of keeping records and making reports is essential to the effective administration and control of any pressure-relieving device program in a process industry. The system should be as simple and clear as possible.

The principal objective in keeping records is to make available the information needed to ensure that the performance of pressure-relieving devices meets the requirements of their various installations. Records may be considered as tools needed to implement the program, and reports may be considered as the means to distribute those tools to all the participants of the program so that they coordinate their work and effectively discharge their responsibilities. In some cases, reports may be retained in files and considered as permanent records.

8.2 The Need to Keep Records

For each pressure-relieving device in service, a complete, permanent record should be kept. The record of each device should include its specification data and a continuously accumulating history of inspection and test results. Some governmental agencies require that certain records and reports be maintained.

The specification record provides the basic information needed to evaluate the adequacy of a device for a given in-

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stallation or for a contemplated change in operating conditions; provides the correct dimensional and material information needed to minimize shop errors and expedite repairs; and provides design information that facilitates the purchase of a similar device and that is required to inventory spare parts. This information allows a pressure-relieving device to be assembled, tested, and exchanged with an identical device on the unit to minimize the time the unit's equipment is unprotected during a scheduled inspection.

Historical records (service records) showing dates and results of inspections are necessary for the follow-up or control phase of the pressure-relieving device program. They enable periodic reviews to determine whether the planned test intervals for a device are being realized. They also provide performance data that helps evaluate the suitability of the device for its particular service, that can indicate problems in the device's design and materials, and that can even indicate a misapplication of the device. It is especially important that the records offer a practical and realistic basis for establishing and maintaining safe and economical inspection intervals for the device.

8.3 Responsibilities

The duties and responsibilities entailed by the various facets of an inspection and testing program for pressure-relieving devices must be clearly defined to avoid confusion and must be explicitly assigned to assure compliance. Some companies assign these duties and responsibilities to equipment inspectors. Others have process unit operators follow up on an established program under the guidance of the engineering-inspection group.

This subsection is not designed to assign responsibilities to any individual. The following outline of duties is meant primarily to facilitate the understanding of how to use the sample record and report forms in the appendix. These duties might be typical of well-designed, pressure-relieving device programs in large process industries. The responsibilities of engineering personnel might include the following:

a. To furnish specifications.
b. To determine allowable pressure settings.
c. To specify test intervals.
d. To record service data.
e. To prepare lists of devices due for inspection.
f. To review performance data and provide engineering services.
g. To purchase replacement valves or spare parts.

The responsibilities of process unit operators might include the following:

a. To initiate work requests.
b. To see that devices are reinstalled in their proper location.
c. To prepare in-service reports.
d. To check for leaking valves.
e. To ensure that the correct block valves (if any) are locked or sealed open or closed as required.
f. To check vents and drains for operability.

The responsibilities of craftsmen might include the following:

a. To perform the mechanical work required to repair, test, reinstall, and attach identification tags to the devices.
b. To maintain a duplicate set of specification records to facilitate repairs.
c. To furnish test reports.
d. To initiate purchase orders for spare parts.

8.4 Sample Record and Report System

The precise recording and reporting format in a pressure-relieving device program is a matter of individual company choice. The forms in the appendix are sample record and report forms. Much of the report writing, record keeping, and inspection scheduling handled by the reports and records could be managed by a properly programmed computer system.

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APPENDIX—SAMPLE RECORD AND REPORT FORMS

The precise recording and reporting format in a pressure-relieving device program is a matter of individual company choice. The following forms are sample record and report forms. Much of the report writing, record keeping, and inspection scheduling handled by the reports and records could be managed by a properly programmed computer system.

The *specification record for a pressure-relieving device* shows a typical permanent card-form record for specifying a pressure relief valve. Its card form provides convenience in handling, filing, and reviewing. This record holds the basic information needed to properly repair or replace the valve.

The *historical record for a pressure-relieving device* shows a typical permanent card-form service record that holds the dates and results of periodic inspections and tests. The information recorded will form a basis for determining test intervals and design changes.

In the record and report program illustrated here, the engineering-inspection group maintains the records and each month informs the operations group responsible for operating the pressure relief valves of the due dates of any work to be done. A report such as that shown in the *inspection and repair work order and report for pressure-relieving devices (with sample data)* is a simple and effective means for initiating inspection, testing, and repair work. Its return to the engineering-inspection group indicates that the operations group responsible for operating the pressure relief valves has taken action. The report should list all the pressure-relieving devices at a given unit to help minimize oversights and clerical work.

When a valve is sent to the shop for inspection, it is inspected and tested by the maintenance group in the "as received" condition. A report such as the *testing report for a pressure-relieving device* is filled out to give the results of this inspection and testing. The report is then sent to the engineering-inspection group, which should copy pertinent information from this report to the historical record and then send this report to the quality-assurance group.

Inspection and testing of a device may lead to its setting and repair by the maintenance group. Orders and records such as the *condition, repair, and setting record for a pressure-relieving device* and the *setting record and repair order for pressure-relieving devices* should be filled out as appropriate.

At the shop, the valve may have a part replaced with a spare part by the maintenance group. In this case, a report is prepared indicating the replacement as well as other basic information on the *condition, repair, and setting record for a pressure-relieving device* form. This report should be sent to the engineering-inspection group (and should arrive there a few days before the "as received" test report).

After a pressure relief valve has been returned to the process unit and installed by the operations group, the authority in the operations group responsible for writing the valve work orders should prepare a report such as the *in-service report for a pressure-relieving device*. This report is filled out to certify that the valve has been reinstalled in its proper location. The report is then sent to the engineering-inspection group. It serves as an independent check on earlier steps and as the final expected report on this particular inspection of the pressure-relieving device.

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SPECIFICATION RECORD FOR A PRESSURE–RELIEVING DEVICE

Device No.	Unit	Location	Set Pressure	Test Interval

Make _____ Style _____

Body and bonnet material _____

Nozzle and disk material _____

Trim material _____

Spring material: ❍ Carbon steel ❍ Alloy

Spring no. _____

Flange sizes

Inlet _____ Outlet _____

Orifice _____ Back pressure _____

Spring set pressure _____

Relieving pressure _____

Normal operating temperature _____

Remarks _____

Maintenance engineer's phone no. _____

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HISTORICAL RECORD FOR A PRESSURE–RELIEVING DEVICE

Device No.	Unit	Location	Set Pressure	Test Interval

Date Tested	Popped	Reset	Disposition	Condition	Repairs	Remarks

API RP*576 92 ■ 0732290 0504057 002 ■

INSPECTION AND REPAIR WORK ORDER AND REPORT FOR PRESSURE-RELIEVING DEVICES (WITH SAMPLE DATA)

To _____
Unit _____
Period _____

All pressure-relieving devices installed on the unit are shown on this list. Those due for testing before expiration of the above period are indicated in the "Due for Test" column by an "X." Please prepare a work request to cover the inspection and repair of the devices indicated and return this work order and report to the engineering-inspection group by _____.

Due for Test	Device No.	Location	Work Requested	Work Request No.
X	FC-100	C-2A compressor discharge		
	FC-101	C-2B compressor discharge		
	FC-102	C-2C compressor discharge		
	FC-103	E-1 tower overhead line		
X	FC-104	E-1 tower bottoms line		
	FC-105	D-35 caustic drum		
	FC-106	J-10A blower discharge		
	FC-107	Rerun tower E-5		
	FC-108	J-63 pump-out		
	FC-109	E-20 exchanger (tube side)		
	FC-110	F-21 drum		
	FC-111	D-37 air drum		
	FC-112	E-2 tower		
	FC-113	E-3 tower		
	FC-114	E-4 tower		
	FC-115	F-1 drum		
	FC-116	E-18 exchanger (tube side)		

Date issued _____ By _____

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TESTING REPORT FOR A PRESSURE–RELIEVING DEVICE

Fill in this report for each device tested and send this report to the quality-assurance group.

Date tested _____
Device no. _____
Unit _____
Location _____
Size _____
Inlet _____
Orifice _____
Outlet _____

Type
Make _____
Style _____
Material _____
Special material _____
Body _____
Trim _____

Fill in blanks below on one side only.

From unit

Date last bench tested _____
Popped at _____
Set pressure _____
Check (dirty pop) pressure _____
If set pressure changed;
new set pressure _____
Test used
Standard _____
Dry seal _____
Disposition
To unit _____
To spare _____
To junk _____
Condition
Leaking _____
Stuck _____
Fouled _____
Corroded _____
Tested by _____

From spare stock or check of new device

Set pressure _____
Check pressure _____

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CONDITION, REPAIR, AND SETTING RECORD FOR A PRESSURE–RELIEVING DEVICE

Manufacturer _____
Figure no. _____
Serial no. _____
Order no. _____
Material of body _____
Trim _____
Spring _____
Bellows _____

Size in _____ inches Rating _____
SIze out _____ inches Rating _____
Spring no. _____
Spring range _____
Orifice no. _____
Valve factor _____
B.D. adj. percent _____

Orig. thickness of body _____
Orig. thickness of outer port _____
Max. pressure in _____ # @ ____ °F
Max. pressure out _____ # @ ____ °F
Bonnet test pressure _____

Sizing cond. _____
Amount _____
Mol. wt. or spec. grav. _____
Rel. temperature _____ °F
Compression factor _____
Percent accumulation _____
Calculation area _____

Set Pressure	Reseat Pressure	Relieving Pressure	Remarks	Test Medium	Dismantled		Disk Stuck		Deposits												Corrosion					Date
									Nozzle				Body				Bonnet									
					N	Y	N	Y	N	L	M	H	N	L	M	H	N	L	M	H	N	L	M	H	Part	

Legend

Deposits and Corrosion
N = None
L = Light
M = Medium
H = Heavy

Parts
B = Bonnet
Be = Bellows
Bo = Body outlet
D = Disk
G = Guide
N = Nozzle
S = Seat
Sp = Spring
St = Stem

Test Medium
A = Air
S = Steam
W = Water
N = Nitrogen
O = Other inert gas

Dismantled/ Disk Stuck
N = No
Y = Yes

Device no. _____
Sheet no. _____
Unit _____
Location _____
Inspection interval _____

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SETTING RECORD AND REPAIR ORDER FOR PRESSURE–RELIEVING DEVICES

To Mr./Ms._____:
Please relieve, dismantle, repair, and reset the following relief valves. Return properly executed original of this form to the engineering-inspection group after completion of work.

Unit _____
Inspector _____
Order no. _____
Date _____
Foreman _____

Legend

Deposits and Corrosion
N = None
L = Light
M = Medium
H = Heavy

Parts
B = Bonnet
Be = Bellows
Bo = Body outlet
D = Disk
G = Guide
N = Nozzle
S = Seat
Sp = Spring
St = Stem

Test Medium
A = Air
S = Steam
W = Water
N = Nitrogen
O = Other inert gas

Dismantled/ Valve Stuck
N = No
Y = Yes

Size	Device No.	Location	Set Pressure	Reseat Pressure	Bonnet Test	No. Notch D. D. Ring	Relieving Pressure	Remarks	Test Medium	Dismantled		Valve Stuck		Deposits												Corrosion					Repairs	Repair Man	Date
														Nozzle				Body				Bonnet											
										N	Y	N	Y	N	L	M	H	N	L	M	H	N	L	M	H	N	L	M	H	Part			

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IN–SERVICE REPORT FOR A PRESSURE–RELIEVING DEVICE

Upon completion of this report, put it in the special envelope provided and send it to the engineering-inspection group.

Device no. _____	Unit _____
Date tested _____	Location _____
Date installed _____	_____

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